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Master Thesis

Enabling flexibility in Robotics Assembly Sequence Generation

by

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Abstract

The increasing complexity of modern manufacturing systems and the rapid evolution of product designs have created a growing demand for flexible and automated assembly planning methods. Traditional assembly planning processes are often performed manually by engineers who interpret Computer Aided Design (CAD) models and define assembly sequences based on experience and design documentation. While such approaches can produce reliable results, they require significant time and effort and are difficult to adapt when product designs change frequently. This research investigates the potential of generating assembly sequences directly from CAD models by analysing geometric relationships between components, thereby reducing reliance on manual interpretation of assembly structures.

To address this challenge, this thesis proposes a geometry-driven assembly sequence generation framework that operates directly on Standard for the Exchange of Product model data (STEP) based CAD models. The framework extracts geometric properties of components, identifies contact relationships between parts, and infers structural dependencies within the assembly. Functional component roles, such as fasteners, are approximated using geometric characteristics including component volume, aspect ratio, and spatial relationships. In addition, collision detection techniques are used to evaluate the geometric feasibility of component removal or insertion operations. The framework was implemented using the Free-CAD platform and Python scripting, enabling direct interaction with CAD geometry for automated reasoning and sequence generation.

Overall, the findings demonstrate that geometric analysis combined with structural dependency reasoning and collision detection can successfully generate feasible assembly sequences that reflect the physical structure of the product. The proposed approach highlights the potential of CAD-based geometric reasoning to support automated assembly planning and strengthen the integration between digital product models and manufacturing process planning in flexible manufacturing systems.

Keywords: assembly planning, CAD-based reasoning, assembly sequence generation, geometric reasoning, collision detection, modular manufacturing systems.

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1 Introduction

The increasing need for electric mobility and energy storage solutions has created greater demand for the production of Battery Module Assemblies, consisting of many individual cylinders or prisms in structural enclosures that provide mechanical support, electrical connectivity, and thermal management [1],[2]. The evolving complexity and diversity of products are further pressuring manufacturers to reduce time-to-market while still delivering a high level of reliability and quality [3]. Automating product assembly is a key enabler for flexible manufacturing, lower costs, and increased production speed [4].

This study proposes an assembly sequence generation method that uses CAD design geometric information to program automated assembly sequences. A primary feature of this methodology is the utilization of the inherent information contained within CAD files to derive assembly knowledge, which would typically require extensive manual production planning [5], [6].

The conventional method of planning assembly in an industrial setting typically occurs after the product's design is complete and mostly relies on experts' knowledge [7]. The engineers will determine how to assemble the product, the assembly sequence, and incorporate them into the robot programs or work instructions. The separate design and process planning phases introduce significant inefficiencies into the industrial workflow when design changes are repeatedly made [5], [8].

An example of product design changes requiring significant changes in process planning and the re-evaluation of assembly methods is the battery module. In order for battery module assembly methods to be effective, configuration changes that include the arrangement of cells in the module, the geometry and volumetric dimensions of battery module enclosures, and the fastening strategy used to hold modules together, will require the re-evaluation and appropriate change of assembly methods [1]. There is a significant incentive to investigate or develop methods that allow for a more automated approach to product geometry interpretation and generate automated assembly sequences with minimal human intervention [6], [9]. Automated assembly methods should reduce manual

effort and enable quick evaluation of the product design, thereby greatly increasing the speed at which the product can move from design to manufacturing [5].

The work presented in this thesis focuses on a computational workflow in which an input feature is represented as a STEP file of a battery module. The model is also parsed to isolate each solid, which may then be analysed to detect functional components such as battery cells, intermediate housing structures, and cover elements. Feature-based reasoning is used to separate cylindrical battery cells from structural members based on geometric attributes, such as surface and volume similarity [10].

Spatial relations and bounding box data are used to identify relative locations from which inferred base components and closing components will be identified. For example, a pre-defined, systematized order of operations that includes rotation, positioning, and insertion along a vertical axis are executed to create the components after they have been determined [9], [11]. The detection of collisions between components and the confirmation that each phase of the operations is geometrically sound demonstrate that the generation of digital product representations into executable assembly knowledge is possible.

This paper explains how semantic representations of geometric shapes can be generated from an engineering perspective for assembly. Despite the presence of exact dimensional and topological specifications in Computer-Generated Images of CAD models, there is no independent method for determining how components can be assembled. To support linking components, there must be an independent logic that applies reasoning to geometrical indicators, thereby producing functional linkages among the different parts. The research provides evidence of the ability to derive a geometrical representation and/or association using rule-based logic and spatial analysis [9], [10].

The major objective of this research project is to develop and evaluate a framework that generates an assembly sequence for a battery module solely from CAD information, and to determine whether the resulting sequence is correct and acceptable for use. To fulfil this objective, four requirements are necessary.

- the framework must be able to identify and classify components based on their geometric features
- the framework must generate a proper assembly order that incorporates both spatial and functional dependencies when assembling components to complete the product.
- the framework must enable assessment of the feasibility of each assembly step using collision detection

- the framework must have an incremental simulation methods, and must also provide documentation creation tools and be integrable into a robotic control system.

All four of these requirements will form the basis for providing a methodology to convert design data into usable knowledge on assembly planning [10], [11].

This thesis presents research questions that define both methodological and practical problems associated with the study of assembly. The main question extends beyond a single geometric feature and examines whether geometric features alone have functional roles, while two related questions examine the reliability of rule-based classification across various design variations and the efficiency of direction-constrained reasoning during assembly [9], [10]. Furthermore, this thesis will examine deficiencies in predefined sequencing logic and highlight areas where more advanced planning strategies, such as optimization or learning-based methods, could provide additional benefits when developing assembly plans. Ultimately, by answering these questions, the thesis will provide an explanation regarding the capabilities and limitations of CAD assembly planning.

Several basic concepts or presumptions support this research and make it feasible in terms of time and resources. It is presumed that the CAD model developed for this research is complete and accurate, with no significant geometric inconsistencies, and that there are no missing or corrupted data points with respect to component identification. Therefore, in this research, the components will also be considered as rigid bodies, and no account will be taken of their deformation, tolerance variation, etc., or any other compliance-type effect. Another assumption underlying this research is that the assembly process for the modular battery will generally use a vertical insertion orientation, as is typically the case in these applications. The actual focus of the study will be on the structural assembly of the modular battery only, with respect to the parameters governing these processes. These assumptions create a simpler problem space for developing sequence generation; however, they still provide basic information from which the assumptions can generate a sequence of operations [7], [12].

Though the simplifying assumptions may have limited our ability to accurately represent assembly reasoning, we still feel that our framework provides a means to capture many core components, such as feature recognition, spatial analysis, and simulation. The framework offers an organized way to validate assembly feasibility before physical prototyping and should be extremely useful in early design phases by identifying potential accessibility or interference issues before committing to large-scale redesigns [10], [11]. The structured format provided by the assembly system of actions and elements clearly shows how knowledge related to assembly can be formalized and easily communicated to other users of digital tools, making it easier to trace the history of an assembly and standardize processes [5], [8].

The purpose of this work is to assess the practicality and potential gains from producing assembly sequence information directly from CAD data for use in manufacturing battery modules. This assessment arises from the necessity for manufacturing processes to be more adaptable to the quick evolution of products. The development and testing of a geometry-based production planner will provide critical insight into how digital representations of a product can help manufacturers greatly reduce manual effort, enable quicker validation, and facilitate the transition to a more intelligent, integrated manufacturing system [3], [4].

This study addresses the computational framework that was developed in Python using Free-CAD to extract the geometric information from the STEP based design models. This method applies the feature recognition and the spatial reasoning information to identify the components and define the assembly precedence relationships based on the assembly planning principles. The collision detection and accessibility checks are combined with this method to ensure the geometric feasibility of the assembly sequence generation. This workflow shows how CAD-based geometric data can be transformed into an fully automated assembly framework.

2 Background and Motivation

Assembly processes have been radically modified by the increasing complexity of the 21st-century manufacturing systems [13], [14]. For instance, throughout history, industrial automation has been driven by an evolution toward more intelligent and connected manufacturing systems [13]. Additionally, as manufacturing products have become more complex and customer demand for customization has increased, assembly processes have had to adapt accordingly [15]. In this regard, robotic assembly has become a key enabling technology for providing repeatability, precision, and continuous operation [16]. Also, while the hardware available for robotic assembly continues to improve rapidly, manual engineering expertise remains the primary means by which assembly processes are developed and configured [17]. The combination of these discrepancies between the physical automation of assembly and the process intelligence that enables it provides the basis for current research in automated assembly planning [18].

Robotic assembly systems were first implemented in manufacturing sectors with high production output and uniformity across items [16]. An example of this is automobile construction, which uses robotics to perform repetitive tasks, such as welding and fastening components into their precise locations [16]. These original assemblies were set up with a strict, predetermined process scale and focused on optimizing a configuration for a single product style. While these original assemblies offered tremendous production benefits, they lacked the ability to accommodate changes to product designs [15]. For these assemblies to produce a different product, all of the following would need to be redone: programming the holding devices and validation. As the manufacturing paradigm changed, the limits of this type of inflexible automated assembly became apparent [15].

Many trends in the marketplace are converging to change how Flexible Manufacturing Systems (FMS) work [15]. These trends include shorter product life cycles, globalization, global competition with other manufacturers, and significant improvements in digital technology. Globalization and innovation also contribute to the continued evolution of FMS in manufacturing citeref16. Manufacturers now recognize that achieving only efficiency does not allow them to compete in the global markets [15]. Therefore, at least two additional important aspects that have led to this significant change are the need for quick, effective

responses to new demand and the need to adapt to meet these demands [15]. Increased awareness of the need for flexibility and adaptability has created a need for new production methods, such as modular production cells and re-configurable tooling, as well as for improved control systems [15]. Furthermore, robot technology has improved, enabling robots to better sense and interact with their environment outside of their control [16]. As a result of these positive developments, the most significant barrier to improving production processes is developing 'assembly sequences', i.e., how to assemble components. This task has remained largely manual [18], [19].

The complexity of planning the assembly of mechanical components stems from the need to consider several geometric aspects, the functional relationships between the individual parts, and the feasibility of the assembly [17]. When determining the sequence for combining component parts, the need to provide access for combining, avoid collisions, and respect dependencies, such as support and alignment requirements, must be considered [18], [20]. Typically, the required evaluations of these considerations are conducted by process engineers, who utilize their prior knowledge and experience through iterative experiments [17]. This method generally yields reliable results; however, it is time-consuming and does not scale well because the product design is often modified. Therefore, researchers are increasingly attempting to develop methods for formalizing assembly knowledge and deriving assembly sequences algorithmically [18], [19].

Digital development, through the creation of new digital collections and materials, has expanded the role of CAD systems in manufacturing workflows from simply providing geometry creation to serving as a source of product information for simulation, analysis, and management throughout a product's life cycle [21]. A particularly well-known file format, STEP, has been developed to provide a standard for representing three-dimensional geometry and topology and to support greater interoperability between software systems [22]. Because highly detailed digital product models are now available, it is possible to extract assembly instructions directly from the design data used to create them [23]. By using this information, it is possible to reduce dependence on manual interpretation of how to assemble a product and to validate its feasibility earlier in the manufacturing preparation process [17].

Due to the rapid expansion of the electric vehicle and renewable energy storage markets, there has been considerable growth in battery assembly production capacity worldwide [24]. Battery modules typically consist of numerous electrochemical cells housed in structural enclosures, with supporting components such as bus bars, cooling components, and protective covers [24]. Most manufacturers assemble their modules using either a layer-by-layer or stack-type configuration; however, because different cell formats have many design differences, such as spacing and enclosure type, manufacturers must continuously adapt their production plans to fit into existing facilities [25]. Therefore, the efficiency of planning for battery module assembly is an extremely important element of being ready to produce assemblies efficiently [24].

Battery module assemblies are characterized by a predominance of vertical insertions or stacking actions [25]. In general, components are added sequentially along an axis of the assembly, usually starting with the structural base, then inserting the cells (or battery cells) into the base’s cavities, and finally applying covers or lids to close the assembly [25]. This uniformity provides benefits in component handling for both assembly and product assembly, as gravity can assist with positioning and stability. From a planning standpoint, having a predominant assembly direction will constrain planning, limiting the search space for possible (or feasible) sequences of component insertion. Therefore, understanding how to incorporate domain-specific characteristics of battery module assembly operations into automated planning frameworks is crucial for practical applications [18].

Although battery modules appear to lend themselves to structured assembly strategies, most industrial procedures are still defined primarily through manual means [17]. Engineers typically use CAD (computer-aided design) models to identify the sizes and relationships among all components of the battery assembly [21]. They then define an assembly sequence based on component access and the tools available during each assembly phase [17]. While CAD and other digital simulation tools can support both visualization and interference checking between components, they typically require the assembly sequence to be defined in advance [20]. The lack of automated reasoning capabilities means that each new module design necessitates a comparable level of planning effort, limiting scalability [18]. Furthermore, manual approaches may overlook subtle geometric constraints that could lead to assembly difficulties later in the product life cycle [19].

One of the key drivers for pursuing this research is to identify a link between the digital representation of a product and executable knowledge for assembling it [23]. When systems can interpret the geometric properties of objects and infer how their components are to be assembled, this reduces the need to develop assembly plans manually, making product production more agile [18]. Furthermore, automatically generating assembly sequences provides designers with an opportunity to assess the feasibility of their designs before producing physical prototypes [20], thereby allowing them to identify assembly problems before making any prototypes [17]. This capability is consistent with the trend toward using model-based system engineering to provide a central source of information about every product from its inception to its disposal as a digital product model [21].

In practical terms, advances in computational geometry, feature recognition, simulation, and other areas provide the foundation for these capabilities [19], [20]. New algorithms can detect and identify specific surface types and patterns, and calculate highly accurate spatial relationships between surfaces. When combined with rule-based reasoning, these algorithms enable classification based on functional intent [19]. Incorporating collision detection and motion feasibility analysis further allows the validation of candidate sequences within a virtual environment [20]. These developments suggest that automated assembly planning is no longer purely conceptual but increasingly feasible with current tools [18].

With an increased focus on digital continuity among manufacturing entities, the motivational element under this umbrella is the need for consistency across product design, process planning, and production execution to reduce errors and maintain traceability [21]. The automated extraction of assembly data from CAD models supports the continuity attribute by reducing the potential for differences compared to a manual interpretation process [23]. Furthermore, the outputs generated by structured planning models can seamlessly interface with other digital platforms to increase interoperability across the system [22]. Although there are significant advantages to automating assembly planning, many obstacles must also be considered [17]. Many assemblies, in reality, do have tolerances, compliance, and complex interactions between parts that cannot be analyzed solely geometrically. For instance, relying solely on shape to determine assembly decisions may not be effective, as factors such as ergonomics, available assembly tools, and the sequencing of the assembly process must also be considered to maximize efficiency. It is apparent that to establish automated methods as decision-support systems rather than substitutes for human judgment, these limitations must be understood [17].

This study draws on three influences: flexible manufacturing needs, new developments in digital engineering, and the acceleration of battery manufacturing technologies. Although assembly planning is a very important and time-consuming physical task [17], there has not been sufficient progress to keep pace with the rapid development of automated equipment [9]. The study aims to improve and unify manufacturing workflows by investigating techniques for generating assembly sequences from CAD data [18], [19]. The reason for this research is to reduce the time engineers spend, allowing more time for design validation prior to production, to achieve the re-usability of knowledge, and to facilitate a greater move towards intelligent, data-based production systems [13]. To achieve this, the research will investigate the opportunities and challenges of generating assembly sequences directly from CAD data through a case study of an assembly around a battery module component.

3 Literature Review

Robotic assembly automation and assembly sequence planning have been central research areas in computer-aided manufacturing for a long time. This literature review is aligned by its methodological evolution and organized functionally, according to the key architectural dimensions to introduce flexibility in geometry-driven assembly planning systems.

The categorisation reflects the mechanisms by which existing research seeks to introduce adaptability, constraint reasoning, and automation into assembly planning. Each category is analyzed to identify its limitations and research gaps.

- Robotic Assembly and Disassembly System Architectures
- Assembly Sequence Planning Methodologies
- CAD-Based Constraint Extraction and Feature Recognition
- Physical Feasibility and Accessibility Reasoning
- Intelligent and Collaborative Robotic Assembly Systems

3.1 Robotics Assembly System Architectures

3.1.1 Classical Robotic Assembly Systems

Robotic assembly systems of the past used a deterministic form of industrial automation. Manufacturing plants typically produced parts with predetermined design and production

specifications; manufacturing was controlled by stable processes and constant cycles. Engineers designed assembly sequences, which are then programmed into a robotic system to provide a predetermined assembly path. As a result, these robotic systems predominantly provided reliability and repeatability rather than adaptive capabilities.

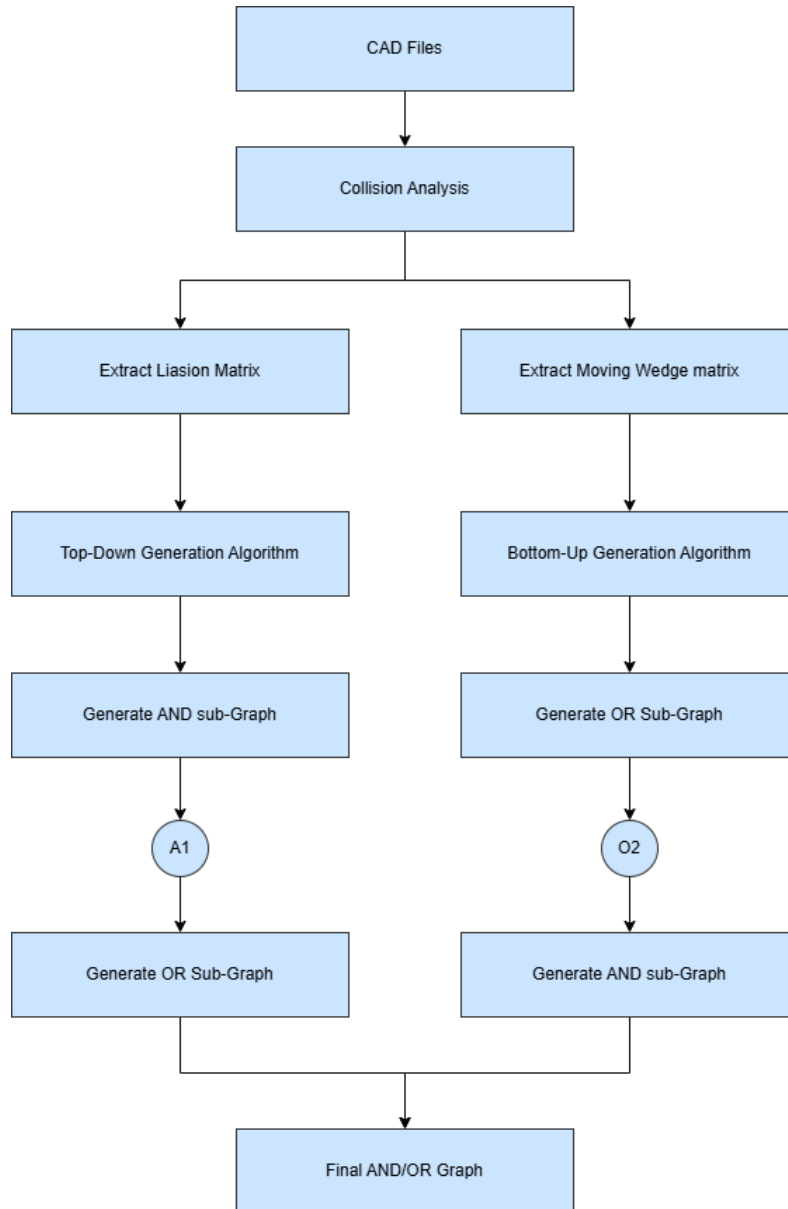


Fig. 3.1: Execution-Architecture of AND/OR Graph of assembly systems [26]

Figure 3.1 presents a complete overview of the assembly and disassembly graph generation process. The flowchart above illustrates how a CAD file serves as the input for the system. Through collision analysis, a set of matrices (Liaison Matrix and Moving Wedge Matrix) is first derived. These matrices are then used to feed two separate generation algorithms

the top-down AOG (Assembly-Oriented Graph) generation algorithm, and the bottom-up AOG generation algorithm.

Based on the output of these algorithms, individual AND/OR sub-graphs are constructed. Finally, by comparing and integrating the results of both approaches, the complete AND/OR graph for the assembly system is obtained.

Zussman and Zhou [27] provided one of the foundational discussions on disassembly modeling, emphasizing structured representations of assembly relationships. Similarly, Santochi et al. [28] presented a comprehensive survey of computer-aided disassembly planning, highlighting that most early systems relied heavily on predefined precedence constraints.

In these types of systems, assembly processes were linked to their physical designs: making changes to a product's shape often meant redrawing all its constraint parameters and sequencing them manually, because there was no automated means of inferring constraints on product assemblies. Additionally, there were very limited abilities to expand or adapt (scale) these types of systems into a new product/assembly.

These types of systems performed well in predictable manufacturing environments; however, they lacked the flexibility to meet the needs of today's fast-paced, dynamic manufacturing environments, which require frequent design changes.

3.1.2 Emergence of Flexible and Modular Automation

As product customization and smaller batch production continue to grow in popularity, manufacturing has shifted towards more flexible automation paradigms. Both modular production lines and re-configurable manufacturing facilities have attracted increased research interest.

Integrated CAD planning frameworks have helped reduce the engineering effort required to assemble a product. Reinhart [29] emphasized the need to integrate digital product data directly into assembly planning. Using CAD-based simulation tools, a sequence can be virtually validated before it is implemented in the physical world.

More recently, the development of digital twin technologies has enabled bidirectional synchronization between the digital and physical systems [30]. These technologies allow for real-time monitoring, predictive analytics, and adaptive control.

While these technologies represent significant advances over traditional planning mechanisms, a key limitation persists for many models: system-wide flexibility does not necessarily translate into planning-level autonomy. Many CAD-integrated systems still re-

quire manual input to produce constraint graphs and/or semantic annotations [31], [32]. Therefore, while digital integration has increased the efficiency of workflows across manufacturing enterprises and product life cycles, many manufacturing models still depend heavily on established structural knowledge.

3.2 Assembly Sequence Planning Methodologies

ASP is the challenge of finding the correct assembly (or disassembly) ordering given constraints, such as geometric constraints and precedence constraints. For an assembly with n components, there are at most n distinct orderings; therefore, exhaustive enumeration for complex products is not feasible.

3.2.1 Rule-Based and Knowledge-Based Approaches

Rule-based planning systems represent one of the earliest formal approaches to ASP. Gu and Yan [31] introduced a CAD-directed automatic assembly sequence planning system in which geometric features extracted from CAD models were analyzed to derive precedence constraints. In this framework, assembly relationships were represented through a set of production rules that captured domain knowledge regarding component interactions and assembly order.

Expert knowledge represented as conditional rules is contained within these systems. Fasteners must be removed before structural members; structural components with blocking relationships are removed in a specific manner, and specific structural components have constraints on tool access prior to removal. The formality of these rules enables the system to generate feasible assembly sequences automatically.

The figure 3.2 shows the execution of the CAD directed automatic ASP. The process begins by constructing a connected graph from the CAD model using binary trees and adjacency matrices, and by calculating the connectivity degree. The assembly is then disassembled into smaller groups based on connectivity analysis. For each of the smaller groups, disassembly sequences are formed by identifying valid removal functions, determining derivatives, and evaluating cost factors. Finally, the disassembly is merged into a single, complete, and realistic assembly sequencing plan for use in robotics. The method described above enables automatic planning of assembly operations using feature data from the CAD model.

The main benefit of rule-based planning systems is their transparency and interpretability. Engineers can track planning decisions through the logic by reviewing the rules used, and

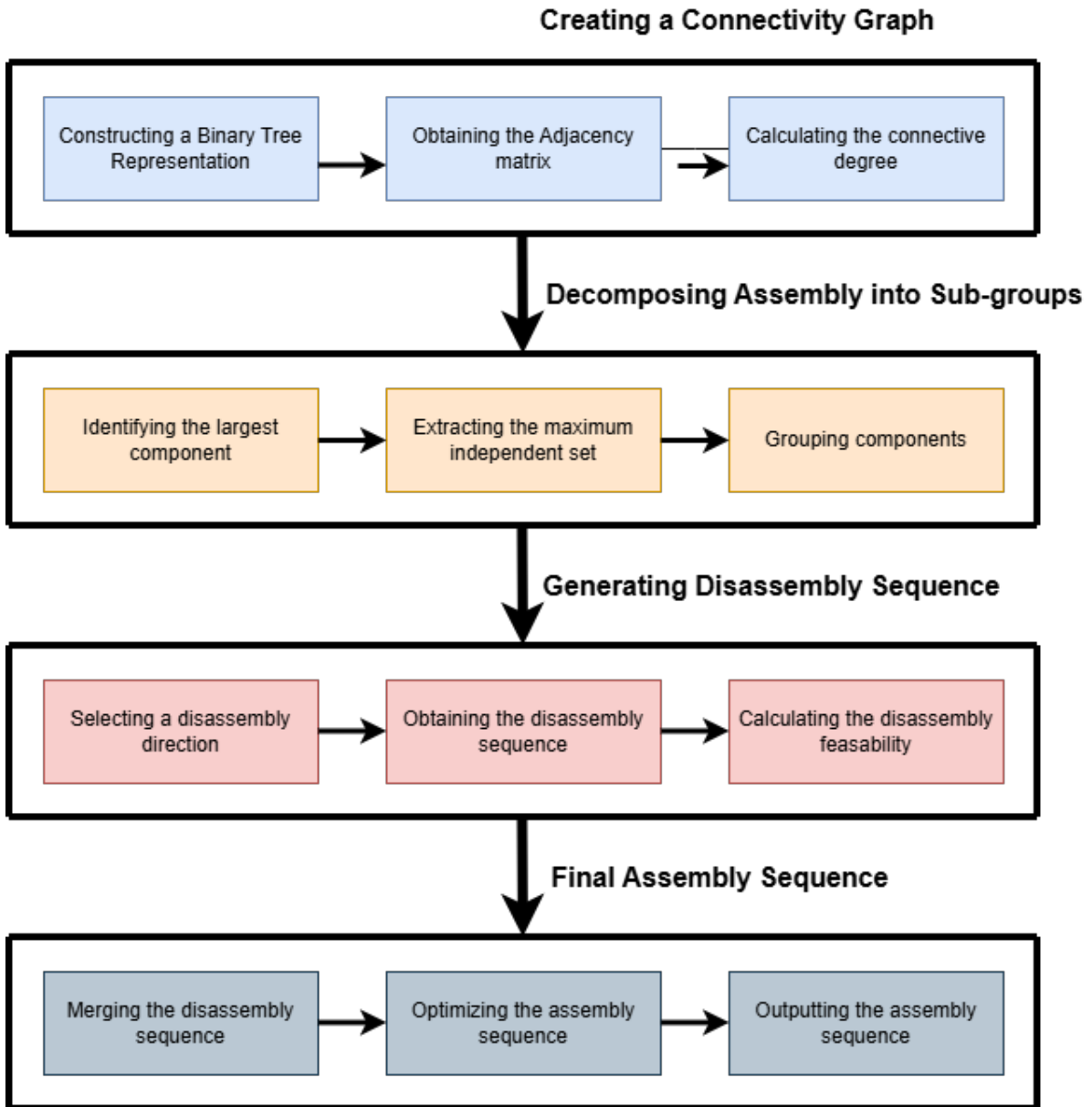


Fig. 3.2: Execution-Architecture of a CAD directed automatic ASP [33]

can change or add rules to meet new production needs, because the logic for the planning decisions is documented in the rules.

Despite their benefits, rule-based planning systems have several disadvantages. Comprehensive rule sets require substantial domain knowledge and extensive manual knowledge engineering to develop. Assemblies become more complicated as the number of possible interactions escalates rapidly with increasing complexity. As a result, rule conflicts arise, and maintaining large numbers of rules becomes difficult. In addition, planning systems struggle with heterogeneous assemblies that use multiple design conventions. Thus, rule-

based planners do not support the scaling of large industrial assemblies with complex geometric interactions.

3.2.2 Graph-Based Planning Approaches

Graph-based planning approaches represent assemblies using structured relational models. In these models, components are represented as nodes, while physical relationships such as contacts, joints, or precedence constraints are represented as edges.

One of the most widely used representations is the AND/OR graph, which allows planners to represent multiple feasible assembly or disassembly paths. Homem de Mello and Sanderson introduced the AND/OR graph representation to model alternative assembly sequences and evaluate feasible operation orders [9].

Graph-based models provide several advantages. They enable systematic exploration of assembly states and support formal reasoning about precedence constraints. In addition, graph structures allow planners to represent alternative assembly strategies and evaluate multiple disassembly paths.

Wilson and Latombe [18] further extended this concept through state-based disassembly planning, where assemblies are represented as sets of valid states connected by transitions corresponding to component removal operations. This formal representation enables rigorous feasibility analysis and supports systematic search for valid sequences.

Despite their advantages, graph-based models typically assume that accurate contact relationships and precedence constraints are already known. The planning algorithm operates on an existing constraint graph rather than discovering constraints autonomously. As a result, significant pre-processing is required to construct the relational representation before planning can begin.

This separation between constraint construction and planning introduces a major bottleneck when dealing with raw CAD assemblies. When constraint extraction requires manual intervention, the benefits of automated planning are significantly reduced.

3.2.3 Optimization-Based Assembly Planning

Assembly planning is a search-based approach that optimizes the assembly sequence by evaluating candidate solutions against an assembly planning metric, such as assembly time, assembly cost, tool changes, ergonomic constraints, or product stability.

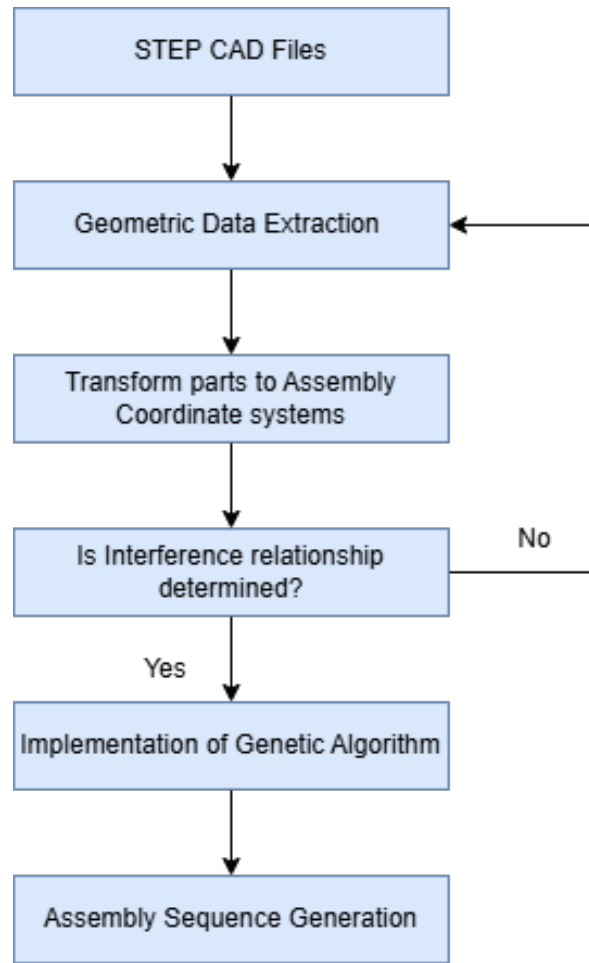


Fig. 3.3: Execution of an Optimization-Based Assembly Sequence Planning [26]

Recently, Ant Colony Optimization (ACO) has been applied to ASP by Yao et al. In their work, artificial agents mimic ant behavior by exploring paths through a constraint graph. Pheromones left behind by agents reinforce previously visited edges, guiding the search towards optimal solutions over time [33].

The given figure 3.3 is the one of some optimization techniques have been applied to the ASP problem that is genetic algorithms, and also other techniques are simulated annealing, and ant colony optimization. Genetic algorithms have been demonstrated to search large solution spaces of STEP-based CAD models by evaluating candidate solutions against geometric and structural constraints derived from CAD assemblies [32].

Optimization-based planners are well-suited for evaluating large search spaces and determining which sequences yield the lowest operational cost or assembly time; they also support multi-objective optimization, thereby enabling planners to achieve multiple performance goals simultaneously.

3.3 CAD-Based Constraint Extraction and Feature Recognition

3.3.1 STEP-Based Assembly Planning

Standardizing CAD file formats, such as STEP (ISO 10303), has greatly enhanced interoperability between design and manufacturing systems. STEP's neutral representation of product shapes and arrangements enables assembly planning algorithms to operate independently of the specific CAD software used to create those designs.

Pan et al. [32] have shown that assembly graphs can be automatically developed by extracting adjacency relationships and geometric constraints from STEP models. Analyzing the topological relationships among components in a given design allows planners to identify possible contact interfaces and establish precedence constraints between them.

While planning systems that are based on STEP's methodology provide an advantage of reducing manual assemblies derived from physical representations of design by establishing geometric relationships, they do not specify the functional roles of each component, i.e., fasteners, load-bearing components, or structural supports, within the STEP representation of the product.

Because of the lack of functional definitions in STEP-based planning systems, additional reasoning mechanisms must still be used to infer the functional relationships between the components of the translated STEP model.

3.3.2 Contact Detection and Geometric Inference

Geometry-driven assembly planning's contact detection heavily relies on the algorithm. They examine a boundary representation (B-rep) model to determine whether two components have intersecting surfaces or physical contact.

Further support for constraint extraction is provided by feature recognition, including geometric features such as holes, slots, and cylinders. Han et al. [34] provided an extensive survey of the various feature recognition methods used in manufacturing.

Tariki et al. [32] provided a comprehensive explanation of how geometric reasoning has been advanced through the introduction of non-interference analysis, which evaluates whether a component can be removed in a specific direction without contacting other parts.

Although they improve the automation for extracting constraints and introduce geometric inference techniques, they are still limited to spatial reasoning. They do not consider a higher-level semantic interpretation of what has been added to that part, meaning they cannot determine if a part is being used as a fastener or as a building support.

3.3.3 Feature Recognition and Semantic Inference

Recognizing features is the process of identifying higher-level geometric features in CAD models. Holes, threads, bosses, and slots can often provide clues about the functional relationships of the items in an assembly.

Although recognition of features allows some inference to build relationships, it can be influenced by the conventions used to create a model. The same functional feature could be modeled differently depending on the CAD strategies used by the modeler.

In addition, there are cases where no direct correlation exists between functional relationships and geometric features. For example, a functional feature might accomplish the goal of connecting parts at a specific location, but does not appear in the connector as part of the feature.

The limitations of feature recognition create points of semantic divergence between its geometry and design function. The geometry in a CAD model expresses its shape and topology, but does not necessarily convey functional meaning.

Bridging this semantic gap is an ongoing challenge for geometric-driven assembly planning.

3.4 Physical Feasibility and Accessibility Reasoning

3.4.1 Collision Detection

Collision detection algorithms are often used to check whether it is geometrically feasible to assemble or disassemble parts. Collision detection algorithms simulate the motion of parts along possible insertion or removal paths, then check for any intersections with the way the parts are configured.

Collision detection algorithms have been thoroughly reviewed in the literature, most notably by Jiménez et al., [11] who conducted a comprehensive survey of algorithms used in computer graphics and robot manipulation that allow planners to determine whether a part can be removed without geometrically interfering with adjacent parts.

Collision detection has only determined the local feasibility of assembly. It has not been determined whether removing parts would affect the structural integrity of any remaining assembly.

3.4.2 Directional Accessibility

Directional accessibility analysis examines whether an item can be removed in a certain direction without being blocked. Tariki and colleagues demonstrated that non-interference analyses can identify possible directions for removing an item by using spatial relationships among items [35].

By providing assurance that candidate sequences/directions will meet required geometric clearance, accessibility analysis can enhance the reliability of automated planning systems.

However, accessibility analyses are limited to local geometric checks and do not account for assembly behavior (e.g., structural support relationships) or the impact of gravity on assembly.

3.4.3 Stability and Support Considerations

Zussman and Zhou emphasized that stability is very important when taking something apart. Removing a component that supports a load may create an unstable situation throughout the assembly and cause it to collapse [27].

Most CAD-based planning systems view assemblies solely from a topological standpoint and do not consider gravity or how one part depends on another for structural support.

An example of this is that many automated planners produce plans for disassembling assemblies that may be geometrically correct but do not make physical sense. A major challenge to integrating structural reasoning into assembly planning remains an area for new research.

3.5 Intelligent and Collaborative Robotic Assembly Systems

To enhance both adaptability and decision-making capabilities, artificial intelligence (AI) techniques have increasingly been incorporated into assembly planning frameworks.

For example, Kousi et al. [36] proposed an AI-based robotic assembly planning framework that combines scheduling algorithms with robotic motion planning to allow robots to dynamically reconfigure the assembly action they are executing in response to changing production requirements.

On the other hand, digital twin technologies enable synchronization between the digital models of a manufacturing system and the physical system. Tao et al. [30] showed that digital twins can support real-time monitoring, predictive maintenance, and adaptive control of machines within a manufacturing environment using the latest technologies.

A third emerging area of research focuses on human-robot collaboration systems. In these systems, an automated planner produces candidate assembly sequences, which are then validated and/or revised by human workers.

While these approaches improve adaptability and robustness in assembly systems, they tend to rely heavily on predefined models of assembly activities and constrain possible assembly activities to those specified in their assembly activity constraints derived from CAD geometry. Therefore, these approaches do not adequately address the issue of discovering constraints autonomously from CAD geometries.

3.6 Comparative Evaluation of Existing Approaches

The literature reveals that existing approaches typically more effective in specific aspects of the assembly planning problem while neglecting others.

Rule-based systems provide transparent reasoning but require extensive manual knowledge engineering. Graph-based models offer structured representation but depend on predefined constraint graphs. Optimization-based planners efficiently explore search spaces but rely on existing constraint models. Collision detection ensures geometric feasibility but does not account for structural stability.

Feature recognition techniques attempt to infer functional relationships but remain sensitive to modeling variations.

Consequently, no existing framework integrates:

- Automatic constraint inference
- Functional component classification
- Physical reasoning (collision, support, gravity)
- Multi-stage planning architectures

This fragmentation defines the boundary of the current state of the art.

3.7 Research Gap Identification

From the comprehensive analysis above, five major limitations can be identified:

- Heavy reliance on predefined assembly constraints
- Lack of automated functional role detection
- Insufficient integration of physical reasoning
- Fragmentation between geometric inference and planning logic
- Absence of end-to-end CAD-native assembly pipelines

Thus, the core research gap can be formulated as:

There is currently no unified geometry-driven assembly and disassembly planning framework capable of automatically inferring structural, functional, and physical constraints directly from raw CAD assemblies and integrating them into a coherent planning architecture.

3.8 Positioning of the Proposed Work

The proposed research addresses these limitations through:

- Automatic contact graph generation from B-rep geometry
- Heuristic fastener detection using geometric metrics
- Directional removal feasibility evaluation
- Support-aware and gravity-aware reasoning logic
- Multi-stage hierarchical refinement
- End-to-end CAD-native implementation

By integrating structural, functional, and physical reasoning within a single pipeline, the proposed framework advances beyond existing state-of-the-art approaches.

Paper	Automatic Constraint Extraction	CAD Integration	Optimization	Physical Reasoning	End-to-End Planning
Gu & Yan (1995)	✓	✓	✗	✗	✗
Pan et al. (2005)	✓	✓	✓	✗	✗
Wilson & Latombe (1994)	✗	✗	✗	✗	✗
Yao et al. (2022)	✗	✓	✓	✗	✗
Münker & Schmitt (2022)	✓	✓	✗	✗	✗
Tariki et al. (2011)	✓	✓	✗	✓	✗
Kousi et al. (2018)	✗	✓	✓	✓	✗
Reinhart (2016)	✓	✓	✗	✗	✗
Proposed Framework	✓	✓	✓	✓	✓

Tab. 3.1: Comparison of Existing Assembly Planning Approaches

The table 3.1 compares various existing approaches to assembly sequence planning against the proposed framework, highlighting key advancements in automation and integration. Earlier works like Gu & Yan [31] and Pan et al. [32] (Iowa State dissertation by Chunxia

Pan on integrating STEP CAD files for automatic planning) achieve automatic constraint extraction and direct CAD integration, with Pan’s method also incorporating genetic algorithm-based optimization, but both lack physical reasoning (e.g., stability, gravity, or force considerations) and true end-to-end planning without manual intervention. Wilson & Latombe [18]) focus on geometric reasoning but offer limited CAD integration, no automatic constraint extraction, and no optimization or physical aspects.

More recent papers such as Yao et al. [33] provide CAD integration and optimization for high-mix low-volume scenarios but miss automatic constraint extraction and physical reasoning; Munker & Schmitt [26] use CAD to generate AND/OR graphs with automatic constraints but lack optimization; Tariki et al. [35] and others add some physical reasoning or partial features, yet none achieve full end-to-end automation. In contrast, the proposed framework stands out by fully satisfying all five criteria—automatic constraint extraction from CAD (likely STEP files), seamless CAD integration, optimization (e.g., via advanced algorithms), inclusion of physical reasoning, and complete end-to-end planning and representing a significant advancement over prior methods, including Pan’s foundational but more limited 2005 contribution.

4 Research Questions

The literature review in Chapter 2 identifies many shortcomings in the approaches to assembly and disassembly planning currently in use. One major limitation of existing approaches to assembly/disassembly planning is their reliance on predefined constraint-based models. Many existing methods also require the manual construction of an assembly graph or the explicit addition of semantic annotations within the CAD model. These limitations make it difficult for existing planning systems to automatically interpret geometric information from CAD assemblies and convert it into executable assembly sequences.

In response to these limitations, the objective of this research is to determine if assembly planning knowledge can be directly derived from the geometric data contained in the CAD model, and to develop a geometric-driven planning framework capable of automating the extraction of structural relationships, evaluating geometric feasibility, and automatically generating executable assembly/disassembly sequences.

To support the efforts of this research, the following research questions will be used:

RQ1: *What are the requirements for an automated workflow transfer for modular assembly systems?*

Modular assembly systems must understand their role for continuously transforming their assembly functions to ensure a sustainable aptitude for automated task design throughout their entire assembly process.

Four capabilities (flexibility, reconfiguration, scaling, and interoperability) in a modular assembly system must be present to support the continuous addition, removal, or substitution of resources and modules with minimal disruption to the overall operation of the modular assembly system.

These four elements will also create a framework for dynamic resource allocation and process adaptability through automated task planning, as well as for surface-level real-time decision-making regarding production conditions, which are subject to irregular change.

Furthermore, this research question aims to gather information on leveraging digital production data, planning algorithms, and knowledge representation models to enable rapid development and enhancements to assembly plans through the application of automated systems. Additionally, this aims to explore the need for standardized interfaces, modular control architectures, and intelligent planning techniques to synchronize assembly modules and maximize coordination and operational efficiency in modular assembly systems.

RQ2: *Is it possible to identify functional components such as fasteners using only geometric characteristics?*

This research question aims to determine whether it is possible to reliably identify functional components, especially fasteners, based on their geometric properties. The ability to detect and classify components based on geometric characteristics is crucial for automatically reasoning about the relationships between assembled components, without having to write semantic annotations (natural-language descriptions of geometric shapes) or label the parts. Geometric characteristics are crucial for automatically reasoning about the relationships between assembled components, and to do so without having to write semantic annotations.

The following sections will discuss the specific geometric features (such as shape descriptions, dimensional parameters, and symmetry) that can be utilized to separate fasteners from other components in a digital assembly model. Similarly, this research question will evaluate the capabilities of feature-extraction methods and geometric pattern recognition techniques to identify different types of fasteners, including screws and locking balls.

This includes scenarios in which components are very similar in appearance to other components (e.g., a screws), or for which the functional attributes of the components cannot be directly inferred from their geometry. The research will also examine how other types of information about assemblies, including information regarding neighboring components and assembly constraints, can be added to, or supplement geometric information and be used to make better decisions regarding whether or not to qualify parts for automated component identification (i.e., fasteners) in a modular assembly environment.

RQ3: *Can geometric reasoning combined with collision detection determine feasible component removal directions?*

This research question aims to investigate whether geometric reasoning, combined with collision detection techniques, can determine valid disassembly (“removal”) directions for

parts in assemblies. Identifying valid disassembly directions is a crucial part of automating the assembly/disassembly planning process. This is especially true in modular manufacturing environments, where individual parts must be handled so that none interfere with the positions of adjacent parts.

In addition, later parts of this study will describe how various geometric reasoning methods that examine spatial relationships, contact surfaces between parts, and their relative orientations may be used to understand the constraints on each part within the assembly structure. Lastly, this study will examine the ability to use a collision detection algorithm to determine whether a given part can be moved or rotated about certain axes without intersecting any adjacent parts.

In addition, this research question will examine whether the use of both geometric analysis and collision detection can provide an organized method for evaluating possible removal directions while ensuring suitable collision-free pathways for extracting components. The subsequent discussion will also include the limitations of using geometric analysis alone, especially in complex assemblies where additional constraints (e.g., mechanisms for fastening parts together, dimensional tolerances, inter-dependencies between assembly components) may affect the ability to identify viable paths for removal directions.

RQ4: *Can integrating structural dependency reasoning with geometric feasibility checks improve automated assembly planning?*

This research question investigates whether combining structural dependency reasoning with geometric feasibility analysis can enhance the effectiveness of automated assembly planning in modular manufacturing systems. Automated assembly planning requires not only an understanding of the spatial relationships between components but also the logical dependencies that determine the correct sequence of assembly or disassembly operations. Integrating these two perspectives may enable more reliable and intelligent planning strategies.

The subsequent sections of this work will examine how structural dependency reasoning can be used to identify relationships between components, such as precedence constraints, fastening relationships, and hierarchical assembly structures. These dependencies define the order in which parts must be assembled or removed, ensuring that the generated assembly plans respect the logical constraints inherent in the product structure.

Furthermore, this research will explore the role of geometric feasibility checks, including geometric reasoning and collision detection, in verifying whether the planned assembly or disassembly actions are physically executable. Geometric analysis enables the evaluation of spatial constraints such as interference, contact surfaces, and feasible motion directions, ensuring that a component can be inserted or removed without colliding with neighboring parts.

By integrating these two approaches, automated assembly planning systems may achieve a more comprehensive understanding of both logical and physical constraints within an assembly. Structural dependency reasoning can determine valid operation sequences, while geometric feasibility checks can validate whether these sequences are practically achievable within the given spatial configuration. This integration has the potential to reduce infeasible plans, improve planning accuracy, and increase the level of automation in modular assembly environments.

Additionally, the research will consider how this combined approach can support dynamic and scalable assembly systems, where modules and components may change over time. A planning framework that incorporates both dependency reasoning and geometric validation could enable automated systems to adapt assembly plans efficiently when product configurations or assembly resources are modified.

Overall, this research question seeks to evaluate whether the integration of structural and geometric analysis can provide a robust foundation for improving automated assembly planning, thereby supporting more flexible, reliable, and intelligent manufacturing systems.

5 Conceptual Framework for Assembly Sequence Generation

5.1 Overview

This chapter presents the conceptual framework used for generating assembly or disassembly sequences from CAD models. The framework aims to transform geometric information contained within a CAD assembly into executable assembly knowledge through a structured reasoning process. Traditional assembly planning approaches often rely on predefined constraint models, liaison graphs, or semantic annotations embedded within CAD systems. However, such information is not always available in raw CAD assemblies and often requires significant manual modeling effort [31], [32].

The proposed framework addresses this limitation by deriving structural relationships directly from geometric information extracted from the CAD model. By analyzing spatial relationships between components and evaluating geometric constraints, the framework is able to generate feasible assembly sequences without requiring predefined assembly graphs or manually defined precedence constraints. This geometry-driven approach is consistent with previous research exploring CAD-based assembly planning and automated sequence generation [26], [31].

5.2 Conceptual Architecture

The conceptual architecture of the proposed framework consists of several interconnected stages that progressively transform CAD geometry into assembly planning knowledge. Similar multi-stage planning pipelines have been widely discussed in assembly sequence planning research.

The main elements of the framework include:

CAD Model Input: A STEP-based CAD assembly is used as the input model containing the geometric representation of all components. The use of standardized CAD formats such as STEP allows structured extraction of geometric and topological information [12].

Geometry Extraction: Individual solid components are extracted from the CAD model and their geometric properties, such as bounding boxes, spatial positions, and volumes, are computed.

Relationship Identification: Spatial relationships between components are analyzed to identify contacts and potential dependencies between parts.

Dependency Representation: The identified relationships are represented in a dependency structure that determines which components must be removed before others.

Sequence Generation: Using the derived dependency information and spatial reasoning, a feasible assembly or disassembly sequence is generated.

This architecture enables the system to convert the raw geometric product data into an assembly knowledge to the modular manufacturing systems, which has been a central objective in computer-aided assembly planning research.

5.3 Geometric Reasoning in Assembly Planning

The conceptual framework relies primarily on geometric reasoning to infer assembly relationships. Geometric properties such as component position, size, and spatial overlap are used to determine interactions between parts.

Bounding box analysis is used to estimate spatial relationships between components, while contact detection provides information about physical connections between parts. These geometric indicators allow the framework to approximate structural dependencies without requiring explicit semantic annotations.

Previous studies have demonstrated that geometric reasoning techniques can effectively support automated assembly planning by extracting structural information directly from CAD models [34]. In addition, collision detection methods can be used to evaluate the feasibility of component movement during assembly or disassembly operations [11].

By combining these geometric indicators, the framework constructs a simplified representation of the assembly structure that can be used for sequence planning.

5.4 Dependency-Based Planning Concept

Once relationships between components are identified, the framework establishes dependency relationships that represent assembly constraints. If a component physically connects or supports another part, it may impose a precedence constraint in the assembly sequence.

Dependency modeling has been widely used in assembly sequence planning research to represent structural relationships between components [9], [10]. In such representations, components whose dependencies have been satisfied can be considered for removal or insertion in the assembly sequence.

Spatial reasoning is then applied to prioritize components that are more accessible or located in higher positions within the assembly. Accessibility analysis and collision detection further ensure that the generated sequence satisfies geometric feasibility constraints [11].

This dependency-based reasoning allows the framework to generate feasible assembly sequences while respecting structural relationships between components.

5.5 Summary

This chapter introduced the conceptual framework used for automated assembly sequence generation from CAD models. The framework focuses on transforming geometric information into planning knowledge through spatial analysis, dependency reasoning, and sequence generation.

By relying on geometric reasoning and contact-based relationship identification, the framework reduces dependence on manually defined constraint models commonly required in traditional assembly planning systems [31], [32].

The conceptual architecture presented in this chapter provides the foundation for the methodology described in the following chapter, where the algorithms and implementation details of the proposed assembly planning system are discussed.

6 Methodology for Automated Assembly Sequence Planning

6.1 Overview of the Methodological Approach

This chapter describes the methodology used to generate automated assembly sequences directly from CAD models. The objective of the proposed methodology is to convert geometric information contained in a CAD assembly into executable assembly knowledge using computational reasoning. Unlike many traditional assembly planning systems that depend on predefined constraint graphs or manually defined assembly relationships, the approach proposed in this research derives assembly relationships directly from geometric properties extracted from the CAD model.

ASP has been widely studied in manufacturing automation. Classical ASP systems require engineers to manually specify precedence constraints, liaison relationships, or assembly rules before an automated planner can generate a valid sequence [31]. Although these approaches can produce feasible assembly plans, they often require significant manual effort and domain expertise. In addition, the manual creation of constraint graphs limits the scalability of such systems when new assemblies are introduced.

The methodology proposed in this research addresses this limitation by developing a geometry-driven planning framework that extracts structural information directly from CAD geometry. The approach integrates geometric reasoning, spatial analysis, dependency modeling, and collision detection to derive feasible assembly sequences. Similar geometric reasoning techniques have been explored in computer-aided assembly planning and robotic manipulation research [9], [34].

The planning system is implemented using the FreeCAD environment combined with Python scripting. FreeCAD provides access to geometric entities stored within CAD assemblies, while Python enables algorithmic analysis of component relationships. The

methodology follows a multi-stage pipeline in which geometric information is progressively analyzed to infer assembly knowledge.

The key principle underlying the proposed methodology is that geometric relationships between components implicitly contain information about assembly structure. By analyzing properties such as spatial position, volume, contact relationships, and accessibility, it is possible to infer precedence constraints and generate feasible assembly sequences.

6.2 System Architecture and Planning Pipeline

The automated assembly planning system follows a structured architecture that converts CAD geometry into assembly sequences through several processing stages. Each stage performs a specific analytical task and contributes to the overall planning process.

The architecture consists of five major modules:

- CAD Geometry Processing Module.
- Geometric Relationship Analysis Module.
- Functional Component Identification Module.
- Dependency and Structural Reasoning Module.
- Sequence Planning and Validation Module.

These modules operate sequentially to transform geometric data into structured assembly knowledge.

The overall workflow of the system is illustrated in Figure 6.1. The first stage of the pipeline imports the CAD model and extracts individual solid components. Each component is analyzed to obtain geometric properties such as bounding box dimensions, spatial coordinates, and volume.

In the second stage, spatial relationships between components are analyzed. Contact detection algorithms determine which components are physically connected. These relationships form the basis for constructing a structural representation of the assembly.

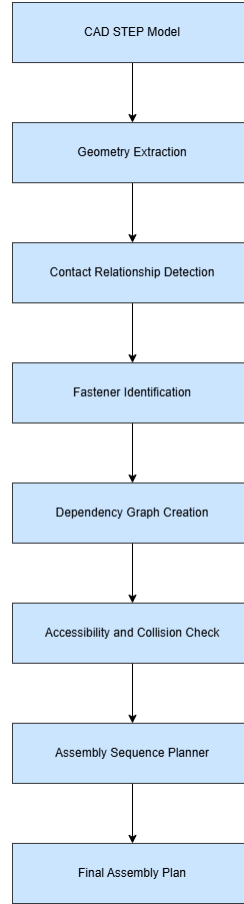


Fig. 6.1: Workflow of the proposed assembly planning methodology

The third stage identifies functional components such as fasteners using geometric heuristics. Fasteners typically exhibit distinctive geometric characteristics, including relatively small volume and elongated shapes.

The fourth stage constructs a dependency graph representing structural relationships between components. This graph defines precedence constraints that determine the order in which components can be removed or assembled.

Finally, the planning module generates a feasible assembly sequence by evaluating component accessibility and verifying that each step satisfies geometric constraints.

This architecture enables the planning system to derive assembly knowledge directly from CAD geometry without requiring predefined semantic annotations.

6.3 Geometric Modeling and Mathematical Representation

In order to perform automated assembly planning, the CAD assembly must be converted into a structured mathematical representation that enables computational reasoning.

Let the assembly consist of a set of components:

$$P = \{p_1, p_2, p_3, \dots, p_n\} \quad (6.1)$$

where each component p_i represents a solid part extracted from the CAD model.

Each component is described by a set of geometric attributes:

$$p_i = (C_i, B_i, V_i) \quad (6.2)$$

where

- C_i represents the geometric center of the component (centroid of the volume)
- B_i represents the bounding box dimensions
- V_i represents the volume of the component

Bounding boxes are used as simplified geometric representations that enable efficient spatial analysis between components.

The spatial relationships between components are represented using a contact graph:

$$G = (P, E) \quad (6.3)$$

where

- P represents the set of components
- E represents set edges graph corresponding to contact relationships between components

Two components p_i and p_j are considered to be in contact if their intersection volume exceeds a small threshold value:

$$\text{Volume}(p_i \cap p_j) > \epsilon \quad (6.4)$$

where ϵ represents a tolerance used to eliminate numerical noise.

Contact graphs have been widely used in assembly planning research because they provide a structured representation of component relationships [31].

In addition to contact relationships, the methodology also considers dependency relationships between components. A dependency relationship exists when the removal of one component requires the prior removal of another component.

These relationships can be represented using a dependency graph:

$$D = (P, R) \quad (6.5)$$

where R represents precedence constraints between components.

Dependency graphs form the basis for assembly sequence generation and have been widely used in assembly planning systems [9].

6.4 Structural Reasoning and Assembly Constraints

The next stage of the methodology focuses on inferring structural relationships between components using geometric reasoning.

Mechanical assemblies often contain components that perform specific functional roles. Fasteners such as screws, bolts, and pins typically connect multiple structural components and must be removed before those components can be separated.

The methodology identifies potential fasteners using geometric heuristics. Components are classified as fasteners if they satisfy several geometric conditions, including small volume relative to other components, elongated shape, and multiple contact relationships with surrounding components.

Automatic fastener detection allows the system to introduce precedence constraints into the dependency graph without requiring explicit semantic annotations.

In addition to fastener relationships, the methodology also considers structural support relationships. In many assemblies, certain components provide a base or structural support for other parts.

Support relationships are identified by analyzing spatial positions of component bounding boxes. If one component is located below another component along the primary assembly direction and their horizontal projections overlap, a support relationship is inferred.

These relationships introduce additional constraints into the dependency graph and prevent the removal of structural base components before supported components have been removed.

Another important aspect of assembly reasoning involves determining the primary assembly direction. Many mechanical assemblies are constructed along a dominant insertion axis.

The methodology determines this direction by analyzing the spatial distribution of components along the three principal axes (X, Y, Z). The axis with the largest cumulative component span is selected as the stacking direction.

This information simplifies accessibility analysis and reduces the search space for sequence planning.

6.5 Assembly Sequence Planning and Feasibility Evaluation

The final stage of the methodology generates a feasible assembly sequence based on the structural relationships inferred from geometric analysis.

The planning algorithm operates iteratively. At each iteration, the system identifies components that satisfy the following conditions:

- The component has no unresolved dependencies in the dependency graph.
- The component is not structurally supporting another component.

- The component can be removed without colliding with remaining parts.

Candidate components are evaluated using accessibility analysis. Accessibility checks determine whether a component can move along the assembly direction without intersecting neighboring components.

Collision detection algorithms simulate component movement and detect geometric intersections between parts. Collision detection is widely used in robotic motion planning and assembly validation systems [11].

If a candidate component passes the accessibility test, it is selected as the next element in the assembly sequence. The component is then removed from the model, and the dependency graph is updated.

The process continues until all components have been removed from the assembly.

The final output of the methodology is a sequence of assembly or disassembly operations that satisfy geometric, structural, and accessibility constraints.

6.6 Methodology Summary

This chapter presented the methodology used to generate automated assembly sequences from CAD models. The proposed approach integrates geometric analysis, contact detection, structural reasoning, and accessibility evaluation to derive assembly knowledge directly from geometric data.

By converting CAD geometry into structured dependency graphs and evaluating component accessibility, the methodology enables automated generation of feasible assembly sequences without relying on manually defined constraint models.

The following chapter presents the implementation of the proposed methodology and demonstrates its application to case study assemblies.

7 Implementation and Case study

7.1 Implementation

The proposed assembly sequence generation framework was implemented using the FreeCAD environment together with Python-based scripting. FreeCAD was selected as the implementation platform because it provides open access to geometric data structures contained within CAD models and supports the analysis of three-dimensional geometry through a programmable interface. The Python API available in FreeCAD enables the development of customized algorithms that can interact directly with CAD geometry, making it possible to perform geometric reasoning and assembly analysis without requiring external software tools.

The implementation begins with importing the assembly model in STEP format. STEP is a widely used neutral CAD file format designed to facilitate interoperability between different CAD systems. By using STEP files as input, the framework ensures that assembly models generated from various CAD design platforms can be analysed without modification.

Once the STEP file is loaded into the FreeCAD environment, a new document is created and the assembly components are inserted into the document structure. The system then performs a recomputation of the document to ensure that all geometric entities are correctly initialized and accessible for analysis. This step allows the framework to retrieve all objects present within the assembly and prepare them for further processing.

The next stage of the implementation focuses on identifying valid solid components within the assembly. CAD assemblies often contain different types of objects, including surfaces, construction geometry, and auxiliary elements that do not represent physical parts. To ensure that only meaningful mechanical components are analysed, the framework filters the imported objects and selects only those that contain valid solid geometries.

Each selected solid object represents a physical component within the assembly. These components are stored within a structured list that will later be used for geometric analysis and sequence planning. By isolating the actual solid components, the system ensures that the planning algorithm operates only on relevant elements of the assembly.

After identifying the components, the framework proceeds to extract geometric properties for each part. These properties include the bounding box dimensions of the component, its spatial centre coordinates, and its overall volume. The bounding box provides a simplified representation of the component geometry and allows the system to perform spatial reasoning without analysing the entire surface geometry of the part.

The bounding box is defined as the smallest rectangular volume that completely encloses the component. From this representation, the system obtains the length of the component along the X, Y, and Z directions. These dimensions are later used to calculate geometric ratios and evaluate spatial relationships between components.

The component centre coordinates are also extracted from the bounding box. These coordinates represent the approximate location of the component within the assembly space and provide a reference point for spatial comparisons.

The volume of each component is another important attribute computed during this stage. The volume provides an indicator of the relative size of components within the assembly. This information is particularly useful when identifying potential fasteners because fasteners are typically smaller than the structural elements they connect.

Once the geometric attributes of all components have been extracted, the framework proceeds to analyse spatial relationships between components. This stage involves detecting contact interactions between parts by performing geometric intersection tests.

Contact detection is implemented by computing the common volume between pairs of components. If two components share a non-zero intersection volume, it indicates that they are in contact or physically connected within the assembly. These contact relationships are recorded in a structured data structure known as a contact map.

The contact map plays a critical role in understanding the structural organization of the assembly. Components that share contact relationships are likely to impose constraints on each other during assembly or disassembly operations. For example, a component that is connected to multiple neighbouring parts may serve as a connector or fastening element.

Using the contact information and geometric attributes of components, the framework attempts to identify potential fasteners. Fastener detection is performed using a set of geometric heuristics derived from typical mechanical design characteristics.

Fasteners usually exhibit certain geometric properties that distinguish them from structural components. These properties include relatively small volumes compared with the overall assembly components, elongated shapes with high aspect ratios, and contact relationships with multiple neighbouring components.

To implement this classification process, the system calculates the median volume of all components within the assembly. Components whose volume falls significantly below this median value are considered candidates for fastener classification.

The aspect ratio of each component is also calculated using its bounding box dimensions. Components that exhibit large ratios between their longest and shortest dimensions are more likely to correspond to elongated shapes typical of screws or bolts.

Additionally, the number of contact relationships for each component is evaluated. Components that interact with multiple neighbouring parts are more likely to serve as connectors between structural elements.

By combining these geometric indicators, the framework identifies components that likely serve as fasteners. These fasteners are then used to establish dependency relationships within the assembly.

The next stage of the implementation involves constructing a dependency structure that represents the order in which components must be assembled or removed. Dependency reasoning is based primarily on the relationships between fasteners and the structural components they connect.

If a component is attached using a fastener, the fastener must be removed before the structural component can be separated from the assembly. This logical constraint is represented as a dependency relationship within the planning model.

Once dependencies are established, the system identifies a dominant stacking axis for the assembly. Many mechanical assemblies exhibit a primary assembly direction along which components are inserted or stacked. Identifying this direction simplifies the planning process by reducing the search space for feasible removal paths.

The stacking axis is determined by analysing the cumulative span of components along the X, Y, and Z directions. The direction with the largest combined span is selected as the primary assembly axis.

After determining the assembly axis, the framework evaluates support relationships between components. Support relationships occur when one component physically supports another component within the assembly structure.

The system evaluates whether a component is positioned above another component along the stacking axis while overlapping in the horizontal plane. If these conditions are satisfied, the lower component is considered to support the upper component.

Support relationships introduce additional constraints into the assembly planning process. Removing a supporting component prematurely may cause instability within the assembly structure.

Finally, the assembly sequence is generated using an iterative planning algorithm. At each iteration, the planner identifies components that have no remaining dependencies and are therefore eligible for removal.

Among the eligible components, the planner evaluates accessibility conditions using a simplified collision detection mechanism. The system checks whether a component can move along the assembly axis without intersecting neighbouring components.

If a component satisfies both dependency and accessibility conditions, it is selected as the next step in the assembly sequence. The component is removed from the assembly model, and the dependency structure is updated accordingly.

This process continues until all components have been removed and the assembly sequence is fully constructed.

The implementation demonstrates how geometric reasoning, dependency analysis, and accessibility evaluation can be integrated within a CAD environment to generate assembly sequences automatically. By operating directly on CAD models, the framework bridges the gap between product design data and manufacturing process planning.

7.2 Description of the Assembly Case Study

To confirm the proposed generation of the assembly sequence within the framework of the research, a case study was conducted using a representative modular robotic assembly system for battery module manufacturing. Battery manufacturing lines using automated assembly consist of multiple automated subsystems that must work together to perform functions such as component handling, cell placement, module assembly, and fastening. The system examined in this research is typical of an automated assembly cell used for battery module manufacturing.

The case discusses an associated assembly system with multiple relationship networks formed from different automatic elements, such as transportation, robotic arms, and a central assembly robot. All of these subsystems perform one of many task functions within the overall manufacturing workflow. The combination of these components forms

a modular manufacturing cell capable of producing the components needed to build a battery module.

The overall workflow of the system begins with battery modules being transported between stations using a conveyor transport system. At specific workstations, robots perform operations such as placing battery cells into the modules, handling components, and performing assembly operations.

The assembly cell used in this study includes the following main components:

- XTS conveyor transport system for module movement
- SCARA robot for pick-and-place operations
- Festo linear robot for assembly and disassembly tasks
- UR5 robot for placing battery cells inside modules
- A future robotic station for screw tightening or locking ball insertion

Each of these subsystems performs a different function within the assembly cell, and together they enable the complete assembly workflow.

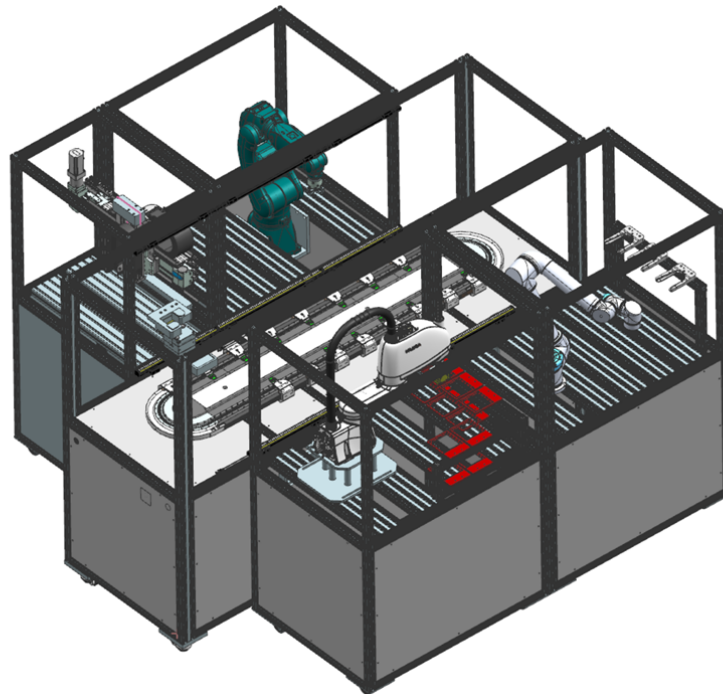


Fig. 7.1: Overview of the robotic battery module assembly system

Figure 7.1 illustrates the general layout of the assembly system.

Battery modules are manufactured using a conveyor system and robots that perform tasks such as inserting cells and placing components. These battery modules consist of many components, including housings, electrical connections, and fastening mechanisms. When assembling these modules, each of the many different robotic systems must complete its separate tasks at precisely timed intervals.

Given this environment, it is possible to evaluate methods for automatically planning assembly sequences for battery modules using geometric analysis of the module components and their relationships. The proposed sequencing method will produce a sequence of assembly events that is commensurate with the physical capabilities of the overall system.

7.2.1 Festo Linear Robot

Festo's linear robot is the key element in this assembly procedure. This robot performs the functions of configuring (assembling) and deconfiguring (disassembling) the battery module.

Linear robots are commonly used in industrial applications because they can execute movements with defined motion in space. Unlike an articulated robot, which uses rotary motion at joints, a linear robot uses only linear (translation) motion along a guide rail. An articulated robot has multiple axes of motion, whereas a linear robot has only one. It therefore requires the item to use linear (translation) motion over a known distance to reach a defined location.

The Festo linear robot in this case study is a multi-axial robot system that uses multiple axes of motion to move a tool or gripper along a predefined motion path. Typically, the Festo linear robot consists of mechanical components, including a structural frame (base), guide rails, motion carriage (the moving portion of the robot), and an actuator mechanism (to provide power).

Together, these components allow the Festo linear robot to move in the X, Y, and Z directions (the three dimensions of a 3D object). By combining the movements of X, Y, and Z, the linear robot can reach different locations within the assembly area to perform various assembly operations on the battery module.

In the battery module assembly operation, the Festo robot handles structural components. Additionally, the Festo linear robot will perform the assembly function of inserting components into the housing and, during disassembly, remove them from the housing.



Fig. 7.2: Festo linear robot used for assembly and disassembly operations [37]

Figure 7.2 illustrates the linear motion system that performs the primary assembly tasks within the robotic cell.

A linear robot interacts with conveyor systems by picking up modules from the transport line and placing them at the assembly station. One advantage of using a linear robot to accomplish these two tasks is that it can achieve high positioning accuracy while working over a relatively large area. As a result, they are commonly used for all repetitive position assembly applications.

The linear robot's geometric arrangement makes it suitable for evaluating automated assembly planning algorithms. It consists of several mechanically connected components, and the components must be assembled in a certain sequence according to constraints.

The proposed framework can determine possible assembly sequences for the robot by examining the geometry of the components in relation to one another.

7.2.2 XTS Conveyor System

The automated XTS conveyor system is used to move battery modules to different assembly cell station locations.

Often, large manufacturing operations have an automated means of moving products between operations, frequently using a conveyor system to provide a consistent and efficient means of moving material in an automated production environment.

Figure 7.3 illustrates the conveyor transport system that moves battery modules between stations within the assembly cell.

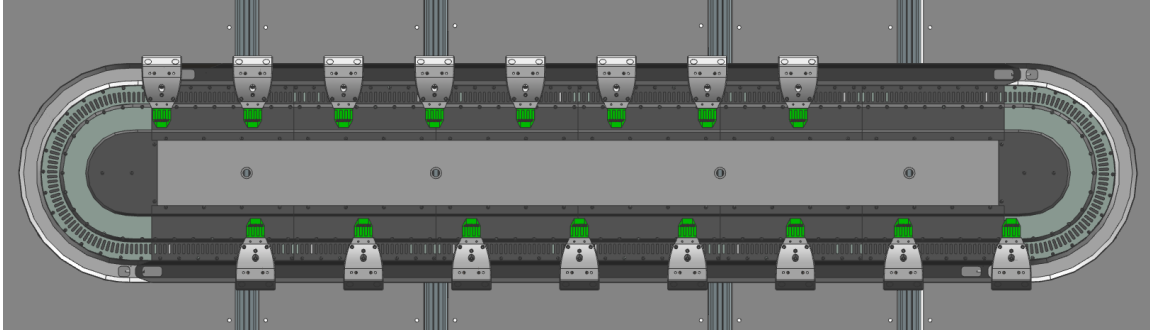


Fig. 7.3: XTS conveyor system used to transport battery modules

The XTS system used in this research is based on a modular transport concept in which individual carriers move along a predefined track structure. Each carrier can transport a battery module from one station to another while maintaining precise positioning control.

A typical conveyor track consists of a series of guide rails that form a continuous transport path. This structure is attached to a framework that provides support to the entire transport process.

The movable carriers that move along the conveyor track transport workpieces between several processing locations. The coordination of this transfer is achieved through sensors and control systems that ensure the carriers arrive at the proper locations at the correct time.

The conveyor provides an essential means for the battery module assembly system to deliver modules from one station to another, where assembly operations will be performed.

An example of this process occurs when the conveyor transports modules to a station for the insertion of battery cells, then transports them to the main assembly area for the final assembly of structural parts.

The utilization of an automated transport system gives individual robots in the assembly cells complete autonomy while maintaining a coordinated production flow between them.

7.2.3 UR5 Robot and Other Robotic Operations

The assembly cell comprises several robotic manipulators that perform different tasks, collectively working with a conveyor system and a linear robot.

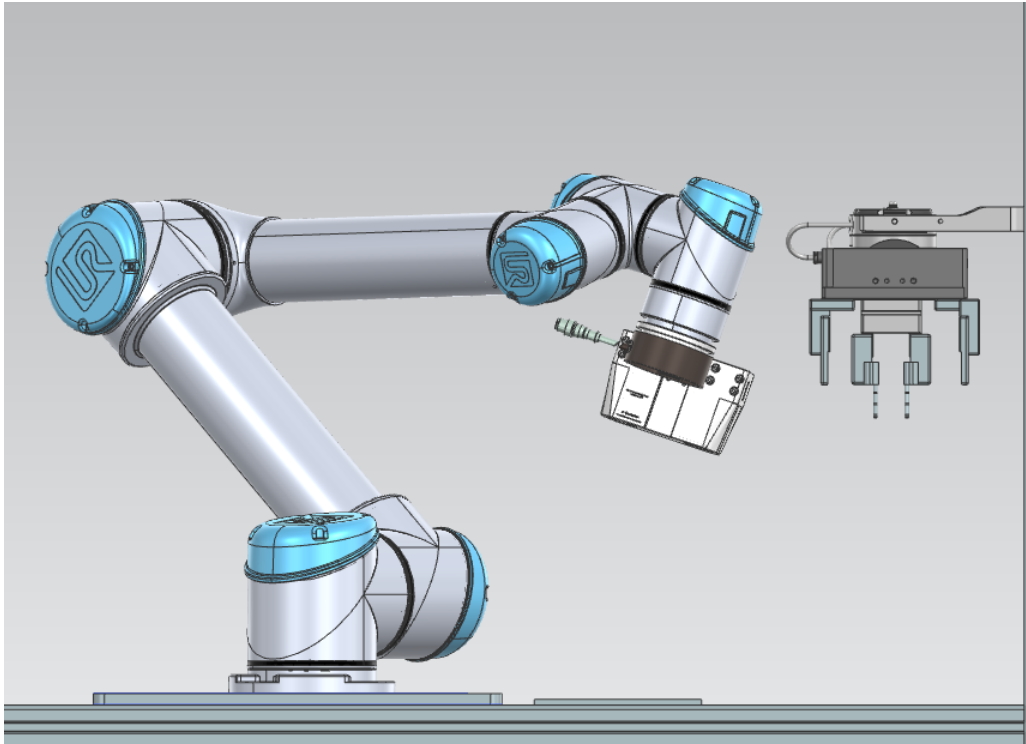


Fig. 7.4: UR5 robot

Figure 7.4 is an example of a UR5 robot manipulator, whereas its task is to install the battery cells into a battery module enclosure within the assembly cell.

The battery cells need to be correctly located within the battery module. This ensures a good electrical connection, such as proper contact between the cell terminals and the battery terminal contacts. The UR5 Collaborative Robot (Cobot) picks up each battery cell from the supply location and places it in its required position inside the module. Cobots like the UR5 are a good choice for these types of work because they are easy to program and teach, and offer flexibility through their articulated structures, allowing them to reach many different positions within their workspace.

Another significant robot within the overall system is a SCARA robot located adjacent to the production line, performing pick-and-place operations between workstations in the production assembly line and supplying materials from the conveyor system. SCARA robots help speed up the assembly process because they can move quickly in the horizontal plane and accurately place components.

The SCARA robot transfers battery module assemblies from the conveyor system to the assembly areas, where they will continue through the production process.

An additional robotic station will be added in the future to perform fastening applications, such as screw tightening and/or inserting locking ball pins. These fastening operations

will be critical to securing the individual components of the battery module after being assembled.

Fasteners such as screws or locking ball pins, hold the structural components of a battery module together during operation.

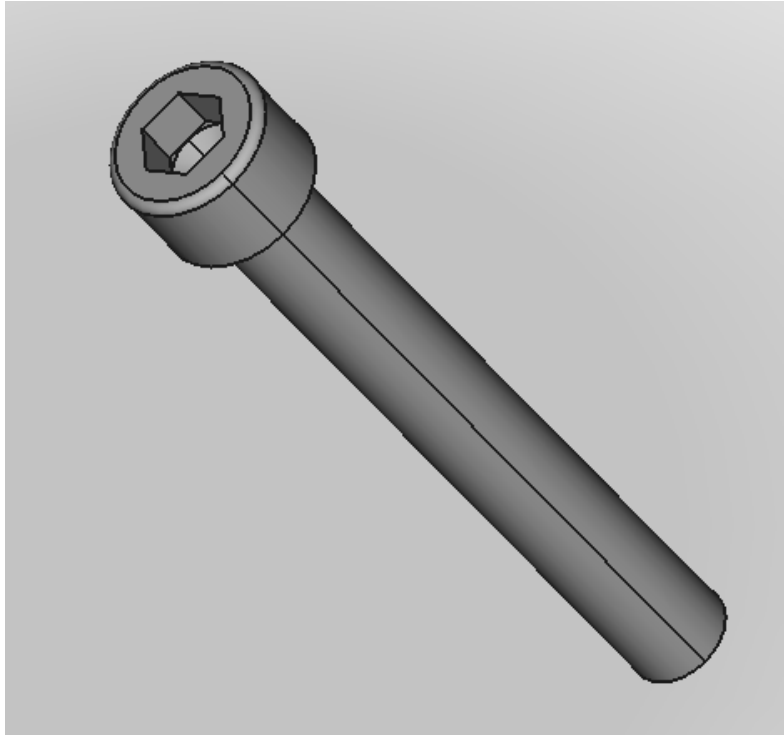


Fig. 7.5: fasteners used in the battery module

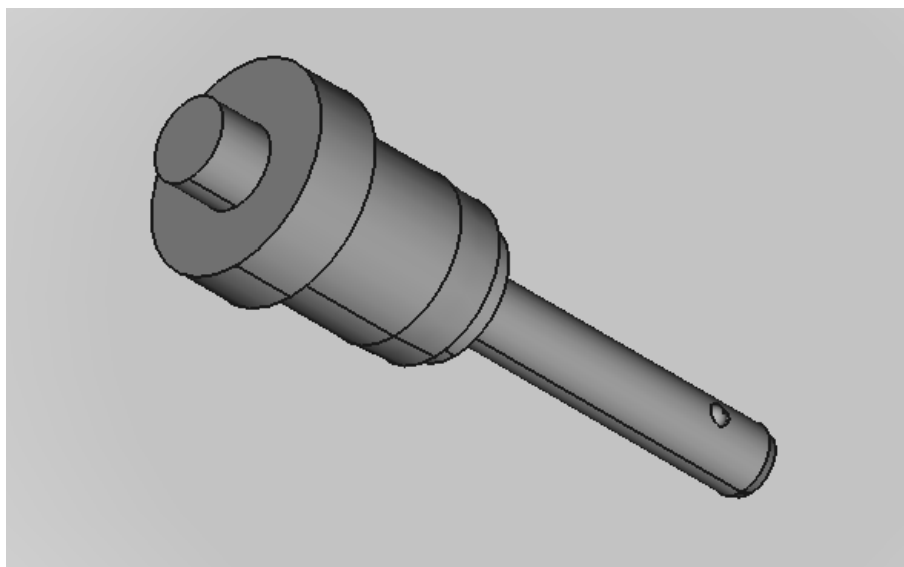


Fig. 7.6: locking ball fasteners used in the battery module

Figure 7.5 and figure 7.6 shows examples of fastening elements that may be used to secure components within the battery module assembly.

In this study, the technology included within the assembly cell is an excellent representation of the types of technologies present in modern automated manufacturing environments. Automated assembly sequence planning methods can therefore be assessed effectively using the technologies described within the assembly cell.

The conclusion of the case study highlights the need for modular assembly equipment, flexible assembly planning systems, and transport systems to support the use of advanced technologies in automated assembly, in line with future automation trends.

7.3 Battery Module Product Description

In order to better illustrate the assembly process considered in this research, it is important to describe the product that is being assembled within the robotic manufacturing system. The product used in the case study is a battery module, which represents a typical subassembly used in electric vehicle battery packs and energy storage systems.

To clearly illustrate the assembly structure of the module, three different representations of the product are shown in Figures 7.7, 7.8, and 7.9. These representations correspond to different stages of the assembly process.

Figure 7.7 shows the fully assembled battery module. In this configuration, all structural components, battery cells, and connectors are installed. This representation illustrates the final state of the product after the assembly process has been completed.

The assembly operations performed on these modules involve several sequential steps, including the placement of battery cells, the insertion of structural components, and the fastening of connectors or locking mechanisms.

The fully assembled module represents the target configuration that must be achieved by the robotic assembly system. All components must be placed correctly and secured to ensure the structural integrity and proper operation of the module.

Figure 7.8 shows a partially assembled module in which the battery cells are visible within the structural housing. This representation highlights the internal arrangement of the cells and demonstrates how they are positioned inside the module.

The placement of battery cells inside the module is performed by the UR5 robot described earlier in this chapter. The robot picks individual cells from a supply station and inserts them into the designated locations within the module housing.

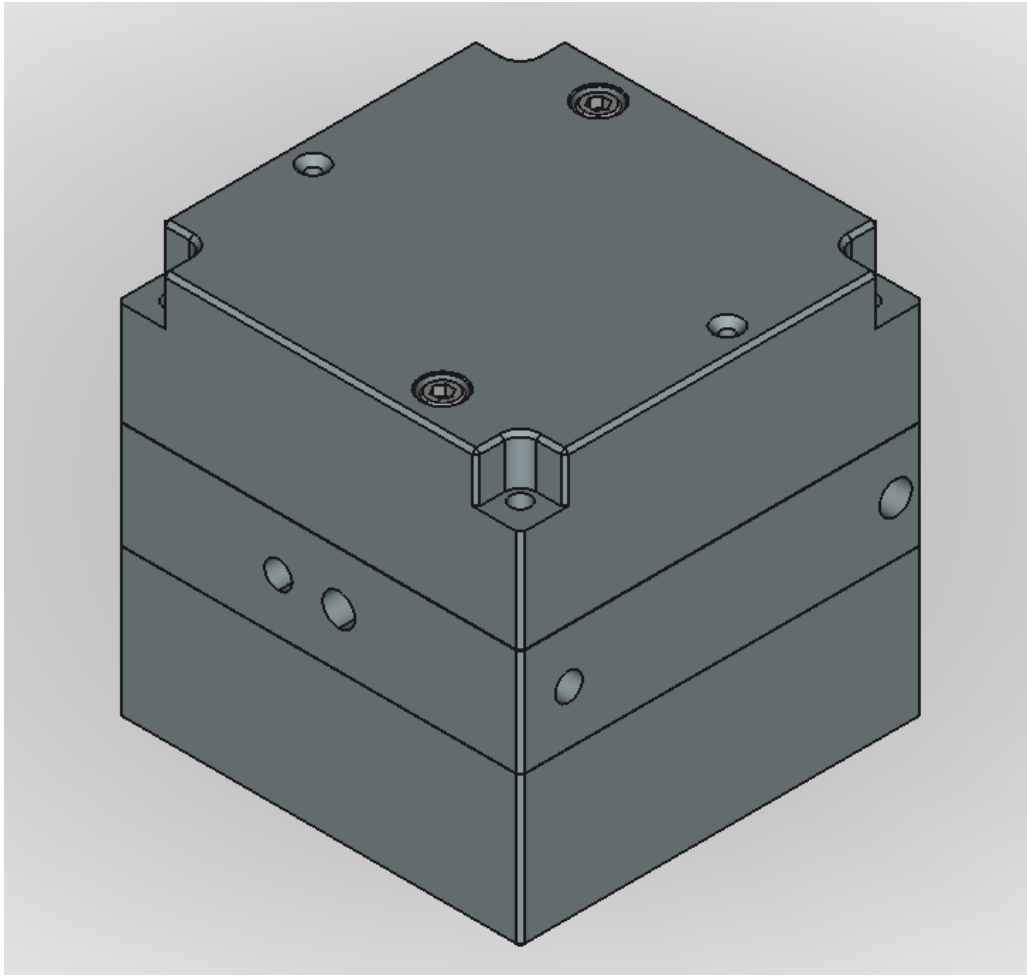


Fig. 7.7: Assembled Battery module

Battery modules serve as intermediary components that contain a large number of battery cells in a mechanically stable arrangement. They provide electrical connections, mechanical support, and thermal management to the individual battery cells. The battery modules are typically assembled by inserting multiple cylindrical or prismatic battery cells into a structural housing and securing them to the housing with mechanical fasteners or connectors.

A battery module typically includes a structural frame, internal support structures, battery cells, and fasteners. The frame serves as the module's external housing, keeping its components together, and the internal support structures maintain proper separation and alignment between the battery cells.

The robotic cell assembly process for the battery module is performed sequentially. The first step is to position and align the module's structural components. The second step is to place the battery cells in their proper locations within the module housing. The third

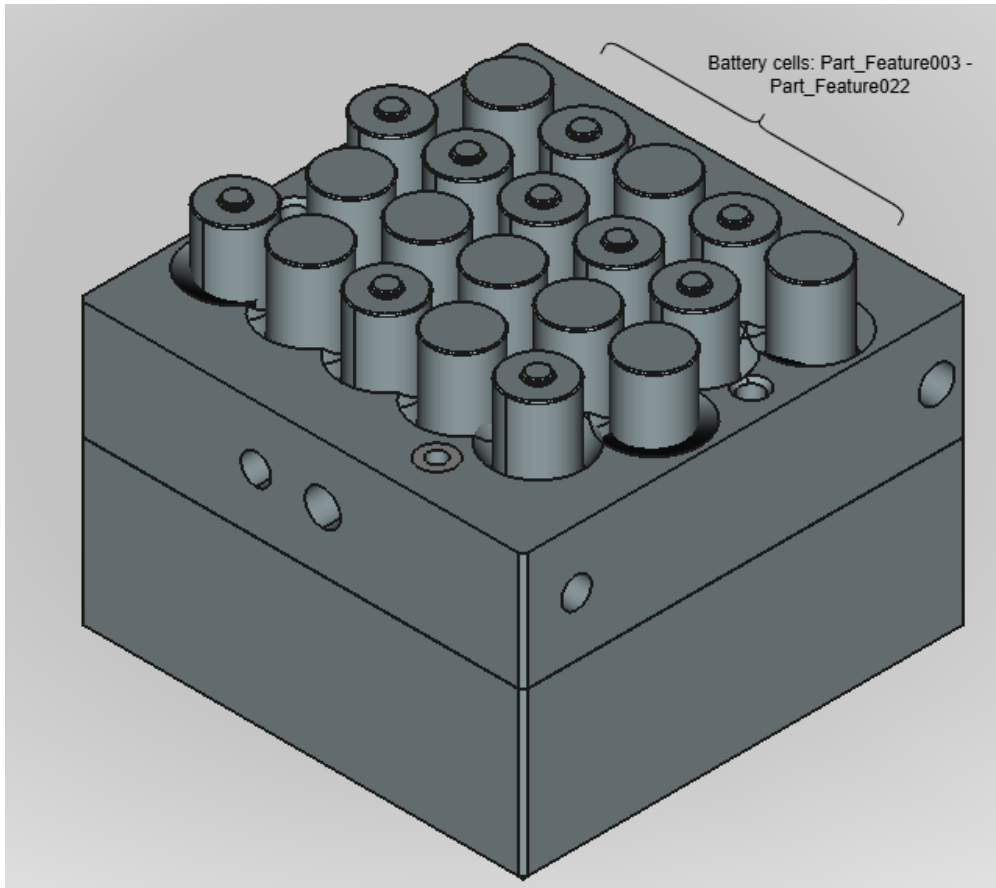


Fig. 7.8: Partially assembled battery module showing battery cells

step is to secure the module's structural components using appropriate fastening devices (screws or locking ball pins).

Finally, Figure 7.9 shows the structural frame of the battery module without the battery cells installed. This representation highlights the mechanical structure that supports the cells and maintains their alignment during operation.

This visual representation will help the researcher in the assembly reasoning process used in this study. The geometric or spatial relationships among the various elements of the structure define how a module may be assembled or dismantled.

Using the geometry of the structural frame and the spatial relationships within each cell, the assembly sequence generation framework will determine the order in which components must be inserted or removed.

These three visual representations of the battery module will therefore provide an excellent visual representation of the product structure and illustrate the assembly operations performed in the robotic manufacturing system.

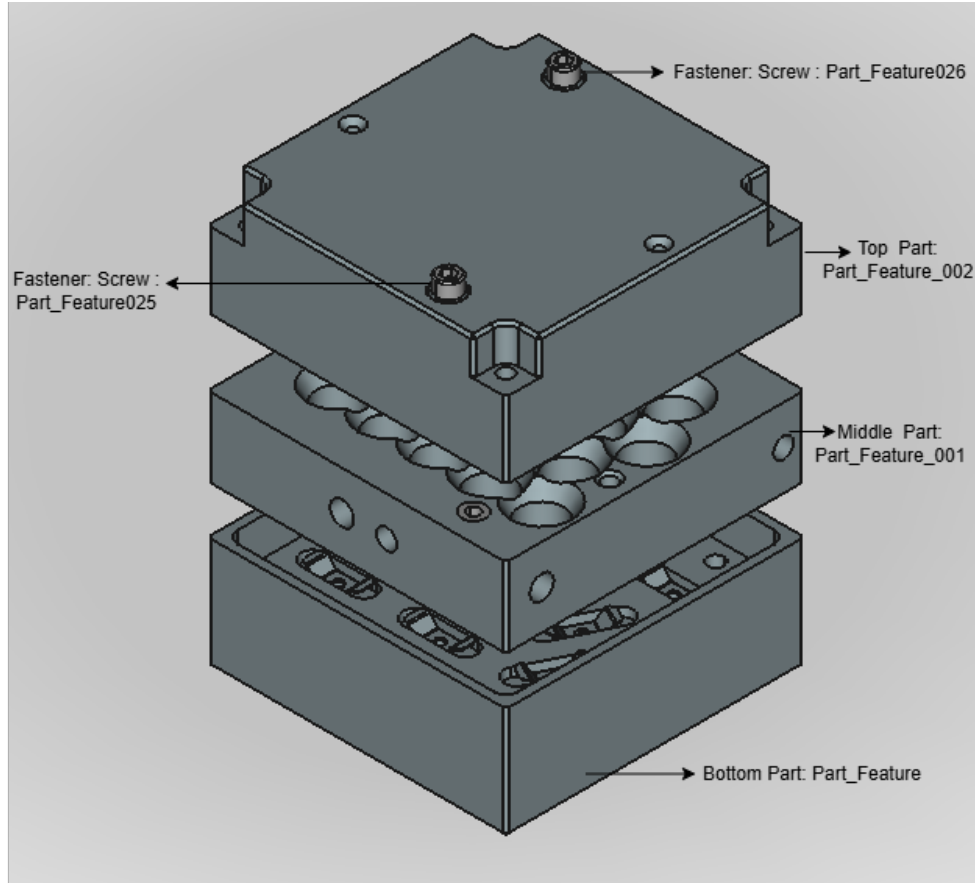


Fig. 7.9: Battery module structural frame without battery cells

7.4 Functional Role Identification Results

When preparing an automated way to create an assembly plan, the first step will be to determine what functions each part (or component) performs in its fit within the assembly. Mechanical assemblies often include many different types of parts: structural components (such as a frame), connectors (to connect different parts of an assembly), and fastening devices (which hold parts of an assembly together). By defining the purpose or function of each component, it is possible to determine the order in which to assemble or disassemble the components.

Most CAD models do not describe the functionality of each component. Rather, the CAD model contains only the geometry of each part. As a consequence, the automated planning system must use geometric information to infer from the model information about function.

In the new framework described in this paper, the process of identifying functional components is based on a combination of heuristic approaches that rely on geometric and spatial relationship analysis of the components. After importing the CAD assembly into

FreeCAD, the framework extracts all solids from the assembly and calculates their geometric attributes.

One geometric attribute analyzed extensively was the bounding box for each component. A bounding box is a rectangular area that outlines a component based on its size and shape, providing a simplified view of the component. By identifying the bounding box dimensions for each component, the framework determined the relative proportions of each component.

Another fundamental characteristic of a component is its volume. Components that are relatively small are typically connectors or fastening devices, while larger components are usually structural components.

Additionally, the framework evaluates the contact relationships (i.e., the number of other components a component contacts) between components. There may be several neighbouring components with which a component needs to make contact. If there are multiple neighbouring components, this may indicate that the component acts as a connector between structural components.

Using the geometric indicators outlined above, the system infers those components that may function as fasteners. Fasteners are typically small in volume, long in length, and have multiple contact relationships.

Several components were successfully identified as potential fasteners using this approach during the case study. All identified components exhibited the aforementioned geometric characteristics and were appropriately classified by the framework.

Components that are typically classified as structural components, such as structural plates and support frames, are also classified as structural components and not as connectors. This allows the planning algorithm to derive dependency relationships between components.

Although the classification methodology is based on heuristic rules, the results indicate that geometric reasoning can provide valuable insights into a component's function.

7.5 Generated Assembly Sequence

The first step in creating an assembly plan is to develop a feasible assembly sequence. This is accomplished by determining the roles of components and their relationships.

The planning algorithm develops assembly sequences iteratively. At each iteration, the system determines which components can be removed while all previously identified de-

dependencies are satisfied. For example, a component that depends on a fastener cannot be removed until the fastener is removed.

Once the structural relationships are determined, the system evaluates geometric accessibility. A geometric accessibility evaluation determines whether a component can be moved to a potential removal location along an assembly axis without colliding with neighbouring components.

To determine whether a collision would occur during component movement along an assembly axis, the system simulates component motion to check for geometric intersections with nearby components.

If the system detects a collision during simulation, the component is deemed inaccessible in that direction, and the planner examines the next viable component.

Using this method, the planning framework creates a sequence of operations that also satisfies the pre-established geometric constraints.

Typically, the sequence begins with removing fasteners or connectors. Once these connectors have been removed, the larger supporting structural components become accessible.

The sequences generated during the case study experiments followed this logical pattern, demonstrating the ability of the framework to infer assembly order from geometric relationships.

7.6 Discussion of Flexibility Achieved

Developing geometry-driven planning systems is motivated by the constructional flexibility and the need to support manufacturing processes that utilize flexible manufacturing systems. Existing assembly planning systems rely on engineer-created assembly sequences or predetermined constructs for each product.

These predetermined assembly sequences are often required to change when there are product changes. Therefore, updating the assembly order can often take time, delaying production.

When developing the framework for this research, it was decided to derive assembly knowledge directly from CAD geometry and utilize it to generate assembly sequences. With this form of assembly planning, the system can change when the structures that require assembly change.

In addition, based on case study experiments, the framework has been validated as a system that can process assembly configurations with little or no change to the planning algorithm when components are added or moved in the CAD model.

As a result of these adaptations, the character of geometry-driven planning has been shown to have potential for flexible manufacturing systems.

7.7 Observations and Insights

The case studies reveal many insights into the potential of geometry-based assembly planning.

First, the geometric relationships of the parts provide important information about the assembly structure. The contact relationships and the relative positions of the components help to identify how the assemblies are built.

Secondly, using multiple methods of reasoning in combination improves the reliability of the planning. Structural dependency analysis will provide logical consistency, while collision detection will provide physical feasibility.

Finally, developing planning algorithms in CAD systems offers many significant advantages: geometry can be accessed directly without conversion. Overall, the case study validates the use of geometry-based assembly planning as a viable approach for developing automated planning systems that support flexible manufacturing.

7.8 Derived Assembly Sequence and Process Plan

The outputs of the proposed assembly sequence generation method are presented using extracted report logs. These results demonstrate how the algorithm processes geometric and relational data to generate a feasible assembly plan. The table 7.1 shows the classification of the given battery modules parts in the Free-CAD report view as labeled in the figures 7.9, 7.8.

Category	Parts
Bottom (Ground)	Part__Feature
Middle	Part__Feature001
Top (Lid)	Part__Feature002
Battery Cells	Part__Feature003 – Part__Feature022
Screws (Fasteners)	Part__Feature023, Part__Feature024, Part__Feature025, Part__Feature026

Tab. 7.1: Component Classification of the Battery Module

7.8.1 Part Information and Volume Analysis

The figure 7.10 shows the number of objects and identifies valid solid parts used for assembly planning. Volume data helps distinguish major structural components from smaller elements like fasteners.

```

06:20:14 Document loaded: AssemblySequenceGen
06:20:14 Number of objects: 43
06:20:14 Real solid parts: 27
06:20:14
Part volumes (mm³):
06:20:14 1 Part__Feature 254901.342
06:20:14 2 Part__Feature001 152957.821
06:20:14 3 Part__Feature002 254229.667
06:20:14 4 Part__Feature003 8084.507
06:20:14 5 Part__Feature004 8084.507
06:20:14 6 Part__Feature005 8084.507
06:20:14 7 Part__Feature006 8084.507
06:20:14 8 Part__Feature007 8084.507
06:20:14 9 Part__Feature008 8084.507
06:20:14 10 Part__Feature009 8084.507
06:20:14 11 Part__Feature010 8084.507
06:20:14 12 Part__Feature011 8084.507
06:20:14 13 Part__Feature012 8084.507
06:20:14 14 Part__Feature013 8084.507
06:20:14 15 Part__Feature014 8084.507
06:20:14 16 Part__Feature015 8084.507
06:20:14 17 Part__Feature016 8084.507
06:20:14 18 Part__Feature017 8084.507
06:20:14 19 Part__Feature018 8084.507
06:20:14 20 Part__Feature019 8084.507
06:20:14 21 Part__Feature020 8084.507
06:20:14 22 Part__Feature021 8084.507
06:20:14 23 Part__Feature022 8084.507
06:20:14 24 Part__Feature023 993.974
06:20:14 25 Part__Feature024 993.974
06:20:14 26 Part__Feature025 993.974
06:20:14 27 Part__Feature026 993.974

```

Fig. 7.10: Extracted part information and volume distribution

7.8.2 Bounding Box Computation

The figure 7.11 shows the computed bounding boxes of all the components. Bounding boxes define the spatial size and position of each part. This information is used for collision checking and placement reasoning.

```

Part bounding boxes:
06:20:14 {'name': 'Part__Feature', 'center': (50.0, 50.0, -21.3), 'size': (100.0, 100.0, 37.4)}
06:20:14 {'name': 'Part__Feature001', 'center': (50.0, 50.0, 10.7), 'size': (100.0, 100.0, 26.4)}
06:20:14 {'name': 'Part__Feature002', 'center': (50.0, 50.0, 42.8), 'size': (100.0, 100.0, 37.4)}
06:20:14 {'name': 'Part__Feature003', 'center': (23.6, 71.25, 13.0), 'size': (14.44, 14.49, 49.85)}
06:20:14 {'name': 'Part__Feature004', 'center': (23.6, 37.25, 13.0), 'size': (14.44, 14.49, 49.85)}
06:20:14 {'name': 'Part__Feature005', 'center': (76.4, 62.75, 13.0), 'size': (14.5, 14.39, 49.85)}
06:20:14 {'name': 'Part__Feature006', 'center': (76.4, 28.75, 13.0), 'size': (14.5, 14.39, 49.85)}
06:20:14 {'name': 'Part__Feature007', 'center': (58.8, 88.25, 13.0), 'size': (14.5, 14.39, 49.85)}
06:20:14 {'name': 'Part__Feature008', 'center': (58.8, 71.25, 13.0), 'size': (14.5, 14.39, 49.85)}
06:20:14 {'name': 'Part__Feature009', 'center': (23.6, 20.25, 13.0), 'size': (14.5, 14.39, 49.85)}
06:20:14 {'name': 'Part__Feature010', 'center': (23.6, 54.25, 13.0), 'size': (14.5, 14.39, 49.85)}
06:20:14 {'name': 'Part__Feature011', 'center': (23.6, 88.25, 13.0), 'size': (14.5, 14.39, 49.85)}
06:20:14 {'name': 'Part__Feature012', 'center': (58.8, 20.25, 13.0), 'size': (14.5, 14.39, 49.85)}
06:20:14 {'name': 'Part__Feature013', 'center': (76.4, 79.75, 13.0), 'size': (14.44, 14.49, 49.85)}
06:20:14 {'name': 'Part__Feature014', 'center': (76.4, 45.75, 13.0), 'size': (14.44, 14.49, 49.85)}
06:20:14 {'name': 'Part__Feature015', 'center': (76.4, 11.75, 13.0), 'size': (14.44, 14.49, 49.85)}
06:20:14 {'name': 'Part__Feature016', 'center': (41.2, 79.75, 13.0), 'size': (14.44, 14.49, 49.85)}
06:20:14 {'name': 'Part__Feature017', 'center': (58.8, 54.25, 13.0), 'size': (14.5, 14.39, 49.85)}
06:20:14 {'name': 'Part__Feature018', 'center': (58.8, 37.25, 13.0), 'size': (14.5, 14.39, 49.85)}
06:20:14 {'name': 'Part__Feature019', 'center': (41.2, 11.75, 13.0), 'size': (14.44, 14.49, 49.85)}
06:20:14 {'name': 'Part__Feature020', 'center': (41.2, 28.75, 13.0), 'size': (14.44, 14.49, 49.85)}
06:20:14 {'name': 'Part__Feature021', 'center': (41.2, 45.75, 13.0), 'size': (14.44, 14.49, 49.85)}
06:20:14 {'name': 'Part__Feature022', 'center': (41.2, 62.75, 13.0), 'size': (14.44, 14.49, 49.85)}
06:20:14 {'name': 'Part__Feature023', 'center': (9.4, 79.75, -17.0), 'size': (7.96, 8.0, 45.0)}
06:20:14 {'name': 'Part__Feature024', 'center': (90.6, 20.25, -17.0), 'size': (7.95, 8.0, 45.0)}
06:20:14 {'name': 'Part__Feature025', 'center': (90.6, 71.26, 38.5), 'size': (8.0, 7.96, 45.02)}
06:20:14 {'name': 'Part__Feature026', 'center': (9.4, 28.69, 38.5), 'size': (8.0, 7.96, 45.02)}

```

Fig. 7.11: Computed bounding boxes of all components

7.8.3 Contact Detection

Contact detection is performed by evaluating geometric intersections and proximity between component bounding volumes. Components sharing common interfaces or overlapping regions are classified as being in contact. This information is used to construct a connectivity graph, where nodes represent parts and edges represent contact relationships. The resulting graph as shown in the figure 7.12 serves as a foundation for dependency extraction and sequence planning.

```

Detected contacts:
06:20:14 Part_Feature001 * Part_Feature023, Part_Feature024
06:20:14 Part_Feature001 * Part_Feature023, Part_Feature024, Part_Feature025, Part_Feature026
06:20:14 Part_Feature002 * Part_Feature003, Part_Feature004, Part_Feature005, Part_Feature006, Part_Feature007, Part_Feature008, Part_Feature009, Part_Feature010, Part_Feature011,
Part_Feature012, Part_Feature013, Part_Feature014, Part_Feature015, Part_Feature016, Part_Feature017, Part_Feature018, Part_Feature019, Part_Feature020, Part_Feature021, Part_Feature022,
Part_Feature025, Part_Feature026
06:20:14 Part_Feature003 * Part_Feature002
06:20:14 Part_Feature004 * Part_Feature002
06:20:14 Part_Feature005 * Part_Feature002
06:20:14 Part_Feature006 * Part_Feature002
06:20:14 Part_Feature007 * Part_Feature002
06:20:14 Part_Feature008 * Part_Feature002
06:20:14 Part_Feature009 * Part_Feature002
06:20:14 Part_Feature010 * Part_Feature002
06:20:14 Part_Feature011 * Part_Feature002
06:20:14 Part_Feature012 * Part_Feature002
06:20:14 Part_Feature013 * Part_Feature002
06:20:14 Part_Feature014 * Part_Feature002
06:20:14 Part_Feature015 * Part_Feature002
06:20:14 Part_Feature016 * Part_Feature002
06:20:14 Part_Feature017 * Part_Feature002
06:20:14 Part_Feature018 * Part_Feature002
06:20:14 Part_Feature019 * Part_Feature002
06:20:14 Part_Feature020 * Part_Feature002
06:20:14 Part_Feature021 * Part_Feature002
06:20:14 Part_Feature022 * Part_Feature002
06:20:14 Part_Feature023 * Part_Feature001
06:20:14 Part_Feature024 * Part_Feature001
06:20:14 Part_Feature025 * Part_Feature001, Part_Feature002
06:20:14 Part_Feature026 * Part_Feature001, Part_Feature002
06:20:14

```

Fig. 7.12: Detected contact relationships between components

7.8.4 Fastener Identification

Fastener identification is carried out by analyzing component geometry and contact patterns to distinguish fastening elements from structural parts. Typically, fasteners exhibit smaller volumes and connect multiple components simultaneously. By recognizing these characteristics, the algorithm classifies specific parts as fasteners shown in the figure 7.13, which introduce critical constraints in the assembly process. These constraints ensure that certain components must be positioned before fastening operations can be performed.

```

Detected fasteners: ['Part_Feature023', 'Part_Feature024', 'Part_Feature025', 'Part_Feature026']

```

Fig. 7.13: Detected fasteners within the assembly

7.8.5 Dependency Generation

The figure 7.14 shows the generated Dependency graph whereas the dependencies define the order in which parts must be assembled or removed. This ensures that the sequence is both feasible and logically consistent.

```

Disassembly dependencies (must remove first):
06:20:14 Part_Feature001 + ['Part_Feature023', 'Part_Feature024']
06:20:14 Part_Feature001 + ['Part_Feature023', 'Part_Feature025', 'Part_Feature024', 'Part_Feature026']
06:20:14 Part_Feature002 + ['Part_Feature025', 'Part_Feature026']
06:20:14
Assembly dependencies (must place BEFORE):
06:20:14 Part_Feature023 + ['Part_Feature', 'Part_Feature001']
06:20:14 Part_Feature024 + ['Part_Feature', 'Part_Feature001']
06:20:14 Part_Feature025 + ['Part_Feature002', 'Part_Feature001']
06:20:14 Part_Feature026 + ['Part_Feature002', 'Part_Feature001']

```

Fig. 7.14: Assembly and disassembly dependencies

7.8.6 Assembly Sequence

The assembly sequence is generated by systematically resolving the dependency graph to produce an ordered list of assembly operations. The sequence incorporates actions such as placement, insertion, and rotation, ensuring that each step satisfies the defined constraints. Structural components are assembled first to establish stability, followed by internal components and fastening operations. The resulting sequence given in the figure 7.15 provides a feasible and logically consistent plan for assembly execution.

```

ASSEMBLY SEQUENCE:
06:20:14
06:20:14  1 ROTATE Part__Feature001
06:20:14  2 PLACE Part__Feature001
06:20:14  3 ROTATE Part__Feature
06:20:14  4 PLACE Part__Feature ON Part__Feature001
06:20:14  5 INSERT Part__Feature023
06:20:14  6 INSERT Part__Feature024
06:20:14  7 ROTATE SUBASSEMBLY
06:20:14  8 PLACE Part__Feature003
06:20:14  9 PLACE Part__Feature004
06:20:14 10 PLACE Part__Feature005
06:20:14 11 PLACE Part__Feature006
06:20:14 12 PLACE Part__Feature007
06:20:14 13 PLACE Part__Feature008
06:20:14 14 PLACE Part__Feature009
06:20:14 15 PLACE Part__Feature010
06:20:14 16 PLACE Part__Feature011
06:20:14 17 PLACE Part__Feature012
06:20:14 18 PLACE Part__Feature013
06:20:14 19 PLACE Part__Feature014
06:20:14 20 PLACE Part__Feature015
06:20:14 21 PLACE Part__Feature016
06:20:14 22 PLACE Part__Feature017
06:20:14 23 PLACE Part__Feature018
06:20:14 24 PLACE Part__Feature019
06:20:14 25 PLACE Part__Feature020
06:20:14 26 PLACE Part__Feature021
06:20:14 27 PLACE Part__Feature022
06:20:14 28 PLACE Part__Feature002
06:20:14 29 INSERT Part__Feature025
06:20:14 30 INSERT Part__Feature026

```

Fig. 7.15: Generated assembly sequences

7.8.7 Process Plan Representation

The generated assembly sequence is further translated into a structured process plan in JSON format. This representation organizes tasks and sub tasks along with corresponding actions required for execution. The figure 7.16 shows the structured format that facilitates integration with automated systems such as robotic manipulators, enabling direct implementation of the assembly process. It also enhances readability and scalability of the planning results. The given figure shows the sample of how the process plan is structured in the JSON format.

```

FULL PROCESS PLAN (JSON):
06:05:01
06:05:01 {
  "tasks": [
    {
      "task": "Assemble Ground and Middle",
      "subtasks": [
        {
          "name": "Turn workpiece",
          "actions": [
            "Activate rotary fixture",
            "Rotate battery module"
          ]
        },
        {
          "name": "Place middle part in assembly region",
          "actions": [
            "Request middle part from transport system",
            "Move robot to assembly station",
            "Place middle part in assembly region"
          ]
        },
        {
          "name": "Turn workpiece",
          "actions": [
            "Activate rotary fixture",
            "Rotate battery module"
          ]
        },
        {
          "name": "Assemble ground with middle",
          "actions": [
            "Request ground part from transport system",
            "Pick ground part",
            "Align ground with middle",
            "Insert ground into middle"
          ]
        },
        {
          "name": "Insert bottom screw Part__Feature023",
          "actions": [
            "Pick screw",
            "Move robot to screw location",
            "Tighten screw"
          ]
        },
        {
          "name": "Insert bottom screw Part__Feature024",
          "actions": [
            "Pick screw",
            "Move robot to screw location",
            "Tighten screw"
          ]
        },
        {
          "name": "Turn workpiece",
          "actions": [
            "Activate rotary fixture",
            "Rotate battery module"
          ]
        }
      ]
    }
  ]
}

```

Fig. 7.16: Generated process plan in JSON format

7.8.8 Summary

The results demonstrate that the proposed method successfully transforms CAD model data into a structured and executable assembly plan. The approach integrates geometric reasoning, contact analysis, and constraint-based sequencing to achieve an efficient assembly workflow. This generated assembly workflow helps to implement the process plan for the modular manufacturing systems to convert the raw geometry driven assembly sequence to an robot executable process plan (JSON).

8 Discussion

This chapter evaluates the outcomes of the research and discusses their implications for automated assembly planning in modular manufacturing systems. The discussion will illustrate how the results presented in this chapter answer the assembly planning research questions presented in earlier sections of this report.

The purpose of this chapter is to provide an interpretation of the results to evaluate their significance in the broader context of automated manufacturing research and to compare the proposed framework and methodology with established assembly planning systems.

The evaluation is structured according to the four research questions established in Chapter 3. The questions examine each of the primary aspects of automated assembly planning, including: system requirements; identification of functional components; establishment, through geometric reasoning, of their respective disassembly directions; and integration of geometric feasibility checks with structural dependency reasons.

This chapter does not provide new empirical data; it presents an interpretation of results derived from the application of the framework and its testing as reported in the prior chapters. Furthermore, the discussion will illustrate how implementing the proposed geometry-driven framework can yield more flexible, adaptable assembly planning capabilities.

In addition to evaluating the research questions, the chapter also includes a comparative analysis of the proposed approach relative to existing assembly planning methods. This comparison helps clarify the strengths and limitations of the proposed framework and identifies areas where further research may be required.

8.1 Evaluation

RQ1:

What are the requirements for an automated control program transfer for modular assembly systems?

The first research question addresses the fundamental requirements necessary for enabling automated control program transfer in modular assembly environments. Modular assembly systems differ significantly from traditional fixed manufacturing systems because they are designed to support rapid configuration changes and dynamic resource allocation.

The analysis conducted in this research indicates that four primary capabilities are essential for enabling automated program transfer within modular assembly systems: flexibility, reconfiguration capability, scalability, and interoperability.

Flexibility refers to the ability of an assembly system to adapt to changes in product design or production requirements without requiring extensive manual reprogramming. The proposed framework contributes to this capability by deriving assembly knowledge directly from CAD geometry rather than relying on manually defined assembly sequences.

Reconfiguration capability enables manufacturing systems to accommodate the addition or removal of modules while maintaining operational continuity. In the proposed framework, assembly planning is based on geometric relationships between components rather than fixed sequence definitions. This allows the system to recompute assembly sequences automatically when changes occur in the assembly structure.

Scalability refers to the ability of the planning framework to operate effectively as the number of components within an assembly increases. Because the proposed system constructs dependency graphs based on geometric relationships, it can theoretically scale to larger assemblies provided that sufficient computational resources are available.

Interoperability is another critical requirement for modular assembly systems. Modern manufacturing environments often involve multiple software tools, control systems, and hardware platforms. By using standardized CAD formats such as STEP, the proposed framework ensures compatibility with widely used engineering tools and facilitates integration within digital manufacturing workflows.

The results of the case study demonstrate that these requirements can be partially addressed through geometry-driven planning techniques. By leveraging digital product data stored in CAD models, the system can automatically generate assembly knowledge that can potentially be transferred to robotic control systems.

However, the study also reveals that automated control program transfer requires additional layers of integration beyond geometric reasoning alone. For example, mapping assembly sequences to robotic motion commands requires coordination between planning algorithms and robotic control systems.

Despite these limitations, the results suggest that CAD-driven planning frameworks provide an important foundation for enabling automated program transfer in modular manufacturing environments.

RQ2:

Is it possible to identify functional components such as fasteners using only geometric characteristics?

The second research question investigates whether geometric information alone is sufficient to identify functional components such as fasteners within CAD assemblies.

The results obtained from the implementation experiments indicate that geometric heuristics can provide useful indicators for identifying fasteners. Components with small volumes, elongated shapes, and multiple contact relationships with surrounding components were successfully detected as potential fasteners by the proposed framework.

These geometric characteristics are consistent with the physical design of many fastening elements such as screws and bolts. As a result, the classification method implemented in this research was able to detect several fastening components within the case study assemblies without relying on predefined labels or semantic annotations.

However, the evaluation also revealed certain limitations associated with purely geometric identification methods. Some components may exhibit geometric properties similar to fasteners while serving different functional roles within the assembly. In such cases, geometric heuristics alone may not be sufficient to determine the exact functional classification.

Another challenge arises when different component types share similar geometric characteristics. For example, cylindrical structural components may appear similar to fastening elements when analyzed solely based on shape and size.

Despite these challenges, the results demonstrate that geometric feature extraction can serve as a valuable tool for approximating functional roles within CAD assemblies. When combined with contextual information such as contact relationships and assembly constraints, geometric reasoning can provide a reliable basis for identifying fastening elements.

These findings suggest that geometry-based component identification can significantly reduce the need for manual annotation of CAD assemblies and support automated reasoning within assembly planning systems.

RQ3:**Can geometric reasoning combined with collision detection determine feasible component removal directions?**

The third research question examines whether geometric analysis combined with collision detection can be used to identify feasible disassembly directions for components within an assembly.

The results obtained from the implementation experiments indicate that geometric reasoning is effective for identifying potential removal directions based on spatial relationships between components.

By analyzing bounding box positions and contact surfaces, the framework can infer the primary assembly direction and identify candidate motion paths for component removal.

Collision detection plays a critical role in validating these candidate directions. Even when geometric reasoning suggests that a component should be removable along a particular axis, surrounding components may physically block the motion.

By performing collision detection during the sequence generation process, the system verifies whether a component can move along the selected direction without intersecting neighboring parts.

The integration of geometric reasoning and collision detection therefore provides a systematic method for identifying feasible removal directions. This approach ensures that generated assembly sequences satisfy both structural and spatial constraints.

However, the evaluation also revealed certain limitations. The current implementation primarily considers linear motion along a dominant assembly axis. In complex assemblies, components may require rotational movements or multi-directional manipulation to be removed successfully.

Additionally, the current collision detection approach does not account for dynamic factors such as flexible components or tool accessibility.

Despite these limitations, the results demonstrate that combining geometric reasoning with collision detection provides an effective strategy for evaluating component removal directions within automated assembly planning systems.

RQ4:**Can integrating structural dependency reasoning with geometric feasibility checks improve automated assembly planning?**

The fourth research question investigates whether combining structural dependency reasoning with geometric feasibility analysis can improve the reliability of automated assembly planning.

The results of the case study analysis strongly support this hypothesis. Structural dependency reasoning enables the planner to identify logical relationships between components, such as fastening connections and support structures. These relationships define precedence constraints that determine the correct order of assembly operations.

Geometric feasibility checks complement this reasoning by verifying whether the planned assembly actions are physically executable. Accessibility analysis and collision detection ensure that components can be inserted or removed without intersecting neighboring parts.

The integration of these two approaches significantly improves the quality of generated assembly sequences. Dependency reasoning alone may produce sequences that are logically correct but physically infeasible. Conversely, geometric feasibility checks alone may identify feasible motions without considering structural dependencies.

By combining these perspectives, the proposed framework achieves a more comprehensive understanding of the assembly structure.

The results demonstrate that this integrated approach reduces the likelihood of generating infeasible assembly plans and improves the overall robustness of the planning system.

8.2 Comparative Analysis with Existing Assembly Planning Approaches

To further evaluate the proposed framework, it is useful to compare its capabilities with those of existing assembly planning methods.

Traditional assembly planning systems typically rely on manually defined assembly graphs or rule-based planning systems. While these approaches can produce accurate assembly sequences, they require significant manual input from engineers.

Optimization-based planning methods attempt to improve sequence efficiency using search algorithms such as genetic algorithms or ant colony optimization. However, these methods still depend on predefined constraint models.

In contrast, the proposed framework derives assembly knowledge directly from CAD geometry. This eliminates the need for manual constraint modeling and allows the system to adapt automatically when assembly structures change.

Another important distinction is the integration of geometric feasibility checks with structural reasoning. Many existing planning systems treat these aspects separately, whereas the proposed framework combines them within a unified planning pipeline.

The comparison of the table 8.1 therefore highlights the main contribution of the proposed approach: enabling assembly planning based directly on geometric information.

Method	Manual Constraints	CAD Geometry Based	Collision Validation	Dependency Reasoning	Automation Level
Rule-based Planning	✓	✗	✗	✓	Low
Graph-based Planning	✓	✗	✗	✓	Medium
Optimization-based Planning	✓	✗	✗	✓	Medium
Feature Recognition Approaches	✗	✓	✗	✗	Medium
Collision-based Planning Methods	✗	✓	✓	✗	Medium
Dependency Graph Methods	✓	✗	✗	✓	Medium
Hybrid Planning Systems	✓	✓	✓	✓	High
Proposed Framework	✗	✓	✓	✓	High

Tab. 8.1: Comparison of Existing Assembly Planning Approaches with the Proposed Framework

8.3 Implications for Modular Assembly Systems

The results of this research have several implications for the design of modular manufacturing systems.

First, the ability to derive assembly knowledge from CAD geometry supports more flexible production environments. When product designs change, assembly sequences can be recomputed automatically rather than manually updated.

Second, the integration of geometric reasoning with structural dependency analysis provides a practical foundation for automated assembly planning systems.

Finally, the use of standardized CAD formats such as STEP facilitates integration with digital manufacturing platforms and supports interoperability between design and production systems.

These capabilities are essential for enabling intelligent manufacturing systems capable of adapting to rapidly changing production requirements.

8.4 Summary

This chapter evaluated the results of the proposed assembly planning framework with respect to the four research questions defined in the study.

The evaluation demonstrates that geometry-driven planning techniques can effectively identify structural relationships within CAD assemblies, detect functional components using geometric heuristics, determine feasible removal directions using collision detection, and generate reliable assembly sequences through integrated structural and geometric reasoning.

The comparative analysis also highlights the advantages of the proposed approach relative to traditional assembly planning methods that rely heavily on manual modeling. Overall, the findings suggest that geometry-driven assembly planning represents a promising direction for future research in intelligent manufacturing systems.

9 Conclusion and Future Work

9.1 Conclusion

This study described methods for generating assembly sequences from digital product models using geometric reasoning techniques. As manufacturing methods become more flexible, there is an increasing need for assemblers to use planning approaches that can respond to changes in product design and evolving production requirements. The majority of assembly methods used today rely heavily on the assembler's experience to guide the creation of manual or automated assembly plans; however, creating automated functionality to generate assembly plans manually can take a significant amount of time and effort. Furthermore, when modifications are made to a design, maintaining a manual assembly generation model can be particularly challenging, as most assembly generation models lack detailed data structures.

This research investigated whether the geometric information contained within CAD assemblies could serve as the primary basis for generating automated assembly sequences through geometric reasoning techniques, using geometric relationships between component parts of an assembly to derive structural dependencies and assess whether assembly operations are physically feasible.

A computational framework in the FreeCAD environment, with Python-based geometric analysis tools, was developed to support this investigation. The framework operates on STEP-based CAD assemblies and performs a series of analysis steps to extract solid components and calculate various geometric properties of each component, including bounding box dimensions, centroid locations, and component volumes. These geometric attributes are then used to identify spatial relationships between components and detect potential contact interactions.

After analyzing these relationships, the framework analyzes the structure of dependencies between components. Connectors between structural components are identified using geometric heuristics based on size, proportion, and contact patterns. This allows the

system to estimate the functional roles of components in the assembly without requiring explicit semantic annotations.

The proposed framework also incorporates geometric feasibility analysis into the planning process. To ensure that an assembly sequence meets both logical dependencies and geometric feasibility, collision detection techniques verify whether a component can move along its intended direction during assembly without intersecting any other parts.

The implementation of the proposed framework and case study shows that the geometric information defined in CAD models can serve as a basis for deriving useful knowledge about how to assemble a product. The framework recognized relationships between components, defined how to fasten them together, and generated an assembly sequence based on the product's structure.

One of the main contributions of this research is to illustrate how digital product models can connect design data with manufacturing planning. By extracting assembly knowledge from CAD geometry, the proposed approach supports the development of more adaptive manufacturing environments in which assembly processes can respond dynamically to design modifications.

The use of modular manufacturing systems is highly valuable in many scenarios, and modular systems require planning systems that enable quick adaptation to changes in modules or product configurations.

The use of geometry-based/reasoning to drive planning processes will provide a strong opportunity to achieve the level of adaptability required to support modular manufacturing.

The research presented in the thesis demonstrates that geometric analysis based on CAD/-CAM can make a major contribution to automated assembly planning. The framework developed in this work is a first step towards a fully automated planning system; however, the results illustrate that geometric reasoning-driven approaches can provide a practical basis for interpreting product structure and creating assembly knowledge.

9.2 Limitations of the Study

There are some limitations to interpreting the findings of this study as they relate to geometry-driven assembly planning.

The first limitation of the proposed framework is the fact that it relies primarily on geometric information to analyze assemblies. Other physical factors involved in manufacturing processes, such as friction, tolerances, material compliance, and component

deformation, may affect the feasibility of the assembly operation; however, these factors are not considered in the implementation of the current framework.

The second limitation of this framework is the development of functional component roles. The framework relies on heuristic geometric rules to determine whether components are fasteners or structural components. Although these heuristics performed well for the case study assemblies, determining functional component definitions will likely be difficult for assemblies with atypical component geometries or highly integrated designs.

The current framework evaluates component accessibility by assuming a dominant assembly direction for component removal. However, many assemblies require more complex movements than a single direction, involving motion across multiple axes or rotation. Addressing these additional aspects of assembly motion will require more advanced reasoning methods to generate the correct assembly motion plans.

Furthermore, the validation of the framework was performed using representative assemblies rather than large industrial-scale products. While the approach is conceptually scalable, additional evaluation on more complex assemblies would be necessary to fully assess its robustness and computational performance.

Despite these limitations, the results of this study demonstrate that geometric reasoning provides a useful starting point for automated assembly planning and highlight several opportunities for further improvement.

9.3 Future Work

The findings of this research suggest several directions for future investigation that could enhance the capabilities of automated assembly planning systems.

The promising area of future research is the integration of assembly sequences and robotic motion planning. Although the particular framework developed in this dissertation produces a feasible assembly sequence, additional algorithms are required to convert it into a format that can execute robot motions. The combination of assembly reasoning with robotic motion planning will enable a more comprehensive automation of assembly processes.

In addition to integrating advanced features, recognition methodologies, and assembly reasoning into robotic motion planning, future research could also apply physics-based simulation techniques to support assembly process planning. By incorporating factors such as friction and contact forces into a simulation model, the system could evaluate the performance of its generated assembly plans with greater confidence than would be possible in a traditional assembly environment. Moreover, the use of physics-based sim-

ulation will allow the system to identify potential stability issues before actual assembly or disassembly activities occur.

Machine learning techniques may also offer new opportunities for improving assembly planning systems. Data-driven approaches could analyse previous assembly operations and learn patterns that assist in predicting feasible assembly strategies. Such techniques could complement geometry-driven reasoning and enable more intelligent planning systems.

Another interesting research direction involves integrating the proposed framework with digital twin technologies. Digital twins provide real-time digital representations of manufacturing systems and enable continuous monitoring of production processes. Linking assembly planning algorithms with digital twin environments could support adaptive planning systems capable of responding dynamically to changes in production conditions.

Finally, further research should investigate the scalability of geometry-driven planning methods when applied to large and complex industrial assemblies. Evaluating the framework on assemblies containing hundreds of components would provide valuable insights into computational efficiency and planning performance. By addressing these research directions, future work can extend the capabilities of geometry-driven assembly planning and contribute to the development of more intelligent and flexible manufacturing systems.

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References

- [1] F. Duffner, N. Kronemeyer, J. Tübke, J. Leker, M. Winter, and R. Schmuch, “Post-lithium-ion battery cell production and its compatibility with lithium-ion cell production infrastructure,” *Nature Energy*, vol. 6, pp. 123–134, 2021.
- [2] D. L. Wood, J. Li, and C. Daniel, “Prospects for reducing the processing cost of lithium ion batteries,” *Journal of Power Sources*, vol. 275, pp. 234–242, 2015.
- [3] H. Lasi, P. Fettke, H.-G. Kemper, T. Feld, and M. Hoffmann, “Industry 4.0,” *Business & Information Systems Engineering*, vol. 6, no. 4, pp. 239–242, 2014.
- [4] M. P. Groover, *Automation, Production Systems, and Computer-Integrated Manufacturing*, 5th ed. Pearson, 2020.
- [5] D. E. Whitney, *Mechanical Assemblies: Their Design, Manufacture, and Role in Product Development*. Oxford University Press, 2004.
- [6] V. Shapiro, “Cad interoperability: The real problems,” *Computer-Aided Design*, vol. 39, no. 7, pp. 595–598, 2007.
- [7] G. Boothroyd, P. Dewhurst, and W. Knight, *Product Design for Manufacture and Assembly*. CRC Press, 2010.
- [8] J. Stark, *Product Lifecycle Management*. Springer, 2015.
- [9] L. S. Homem de Mello and A. C. Sanderson, “And/or graph representation of assembly plans,” *IEEE Transactions on Robotics and Automation*, vol. 7, no. 2, pp. 188–199, 1991.
- [10] A. Bourjault, “Contribution to a methodological approach of automated assembly,” PhD thesis, University of Franche-Comté, 1984.
- [11] P. Jiménez, F. Thomas, and C. Torras, “3d collision detection: A survey,” *Computers & Graphics*, vol. 25, no. 2, pp. 269–285, 2001.
- [12] International Organization for Standardization, *Iso 10303 – industrial automation systems and integration – product data representation and exchange (step)*, ap242, 2014.
- [13] H. Kagermann, W. Wahlster, and J. Helbig, “Recommendations for implementing the strategic initiative industrie 4.0,” *Final Report of the Industrie 4.0 Working Group*, 2013.
- [14] L. Monostori, “Cyber-physical production systems: Roots, expectations and rd challenges,” *Procedia CIRP*, vol. 17, pp. 9–13, 2014.
- [15] Y. Koren, U. Heisel, F. Jovane, T. Moriwaki, G. Pritschow, A. Ulsoy, and H. Van Brussel, “Reconfigurable manufacturing systems,” *CIRP Annals*, vol. 48, no. 2, pp. 527–540, 1999.
- [16] B. Siciliano and O. Khatib, *Springer Handbook of Robotics*. Springer, 2016.
- [17] R. Bogue, “Robots in the automotive industry: A review,” *Industrial Robot*, vol. 39, no. 1, pp. 52–57, 2012.
- [18] R. H. Wilson and J.-C. Latombe, “Geometric reasoning about mechanical assembly,” *Artificial Intelligence*, vol. 71, no. 2, pp. 371–396, 1994.

- [19] T. L. de Fazio and D. E. Whitney, "Simplified generation of all mechanical assembly sequences," *IEEE Journal of Robotics and Automation*, vol. 3, no. 6, pp. 640–658, 1987.
- [20] M. Abramovici, *Smart Product Engineering*. Springer, 2013.
- [21] International Organization for Standardization, *Iso 10303-242 industrial automation systems and integration – product data representation and exchange*, 2014.
- [22] M. J. Pratt, "Introduction to iso 10303 – the step standard," *Journal of Computing and Information Science in Engineering*, vol. 1, no. 1, pp. 102–103, 2001.
- [23] N. Nitta, F. Wu, J. T. Lee, and G. Yushin, "Li-ion battery materials: Present and future," *Materials Today*, vol. 18, no. 5, pp. 252–264, 2015.
- [24] J. Li, Z. Du, and R. E. Ruther, "Battery manufacturing processes and design considerations," *Journal of Power Sources*, vol. 412, pp. 1–13, 2019.
- [25] J. Canny, "Collision detection for moving polyhedra," *IEEE Transactions on Pattern Analysis and Machine Intelligence*, vol. 8, no. 2, pp. 200–209, 1986.
- [26] T. Munker and R. Schmitt, "Cad-based and/or graph generation algorithms in assembly planning," *Procedia CIRP*, vol. 107, pp. 1024–1029, 2022.
- [27] E. Zussman and M. Zhou, "A methodology for modeling and analysis of disassembly processes," *IEEE Transactions on Robotics and Automation*, vol. 15, no. 1, pp. 190–199, 1999.
- [28] M. Santochi, G. Dini, and F. Failli, "Computer aided disassembly planning: State of the art and perspectives," *CIRP Annals*, vol. 51, no. 2, pp. 507–529, 2002.
- [29] G. Reinhart, "Cad-based automated assembly planning for variant-rich products," *CIRP Annals*, vol. 65, no. 1, pp. 25–28, 2016.
- [30] F. Tao, H. Zhang, A. Liu, and A. Nee, "Digital twin driven smart manufacturing," *Journal of Manufacturing Systems*, vol. 48, pp. 157–169, 2018.
- [31] P. Gu and X. Yan, "Cad-directed automatic assembly sequence planning," *Computer-Aided Design*, vol. 27, no. 3, pp. 191–205, 1995.
- [32] J. Pan, S. Smith, and G. Smith, "Automatic assembly sequence planning from step models using genetic algorithms," *International Journal of Production Research*, vol. 43, no. 16, pp. 3451–3470, 2005.
- [33] X. Yao et al., "Computer-aided assembly sequence planning using ant colony optimization," *Procedia CIRP*, vol. 107, pp. 1030–1035, 2022.
- [34] J. Han, M. Pratt, and W. C. Regli, "Manufacturing feature recognition from solid models: A status report," *IEEE Transactions on Robotics and Automation*, vol. 16, no. 6, pp. 782–796, 2000.
- [35] M. Tariki, J. Urbanic, and H. ElMaraghy, "Non-interference analysis for automated disassembly planning," *International Journal of Production Research*, vol. 49, no. 14, pp. 4305–4325, 2011.
- [36] N. Kousi, C. Stentzel, G. Michalos, and S. Makris, "Robot-based flexible assembly planning using artificial intelligence," *Procedia CIRP*, vol. 67, pp. 132–137, 2018.
- [37] Festo SE & Co. KG, "Handling systems – individual solutions," 2026. [Online]. Available: https://www.festo.com/de/de/e/loesungen/individuelle-loesungen/handhabungssysteme-id_4462/.